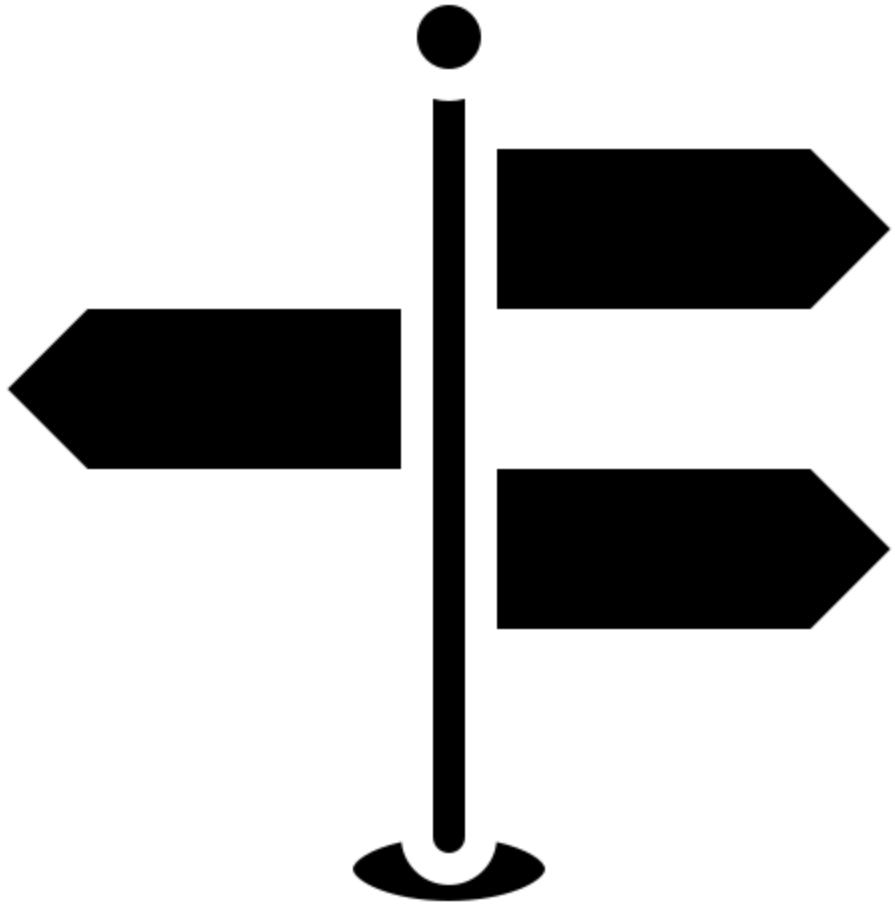
Public Cloud Services
Databases and Storage

Learning Objectives

- Students understand the difference between blob and file storage.
- Students can set up blob and file storage in the cloud.
- Students understand the difference between databases and the purposes of blob and file storage.



Agenda

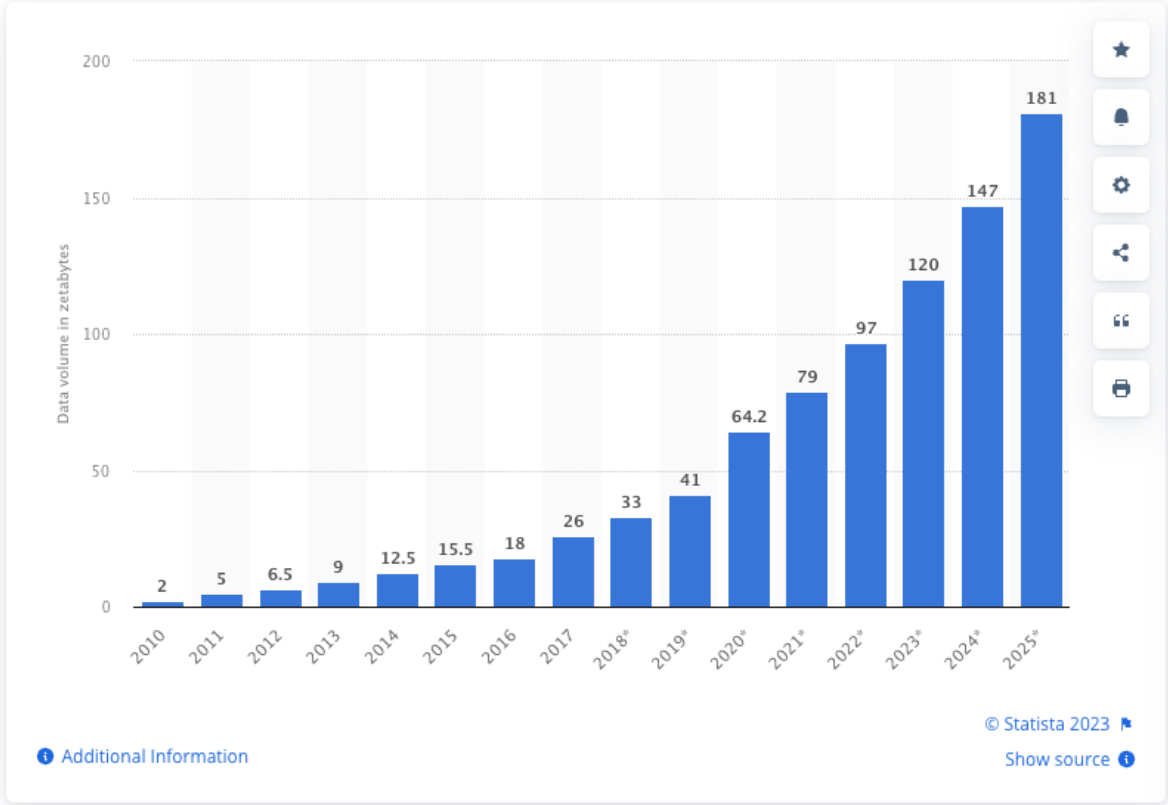
- Storage Concepts
- File Storage
- Block Storage
- Objekt / Blob Storage

Motivation

1. 262 Pages = 1MB
2. 536 Books = 1GB
3. 549755 Books = 1TB
4. 192 Library of Congress = 1PB

– → Non-linear increase in storage volume per year

– Source: https://www.bookpedia.de/buecher/Wieviel_Information_gibt_es_in_der_Welt



Storage Concepts

File Storage

- Access is provided, for example, via a POSIX-compatible file system over the network.
- Files can be written partially.
- Network protocols such as NFS allow multiple clients to access the same files.

Block Storage

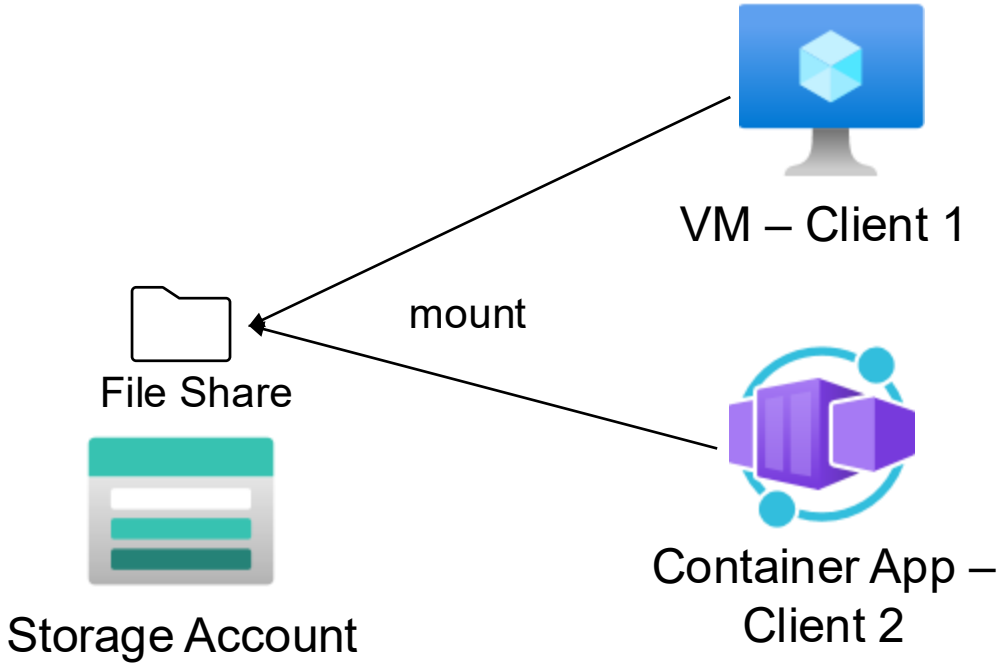
- Accesses are made directly to the device (e.g., disk) attached to a VM.
- Writing is done in blocks, e.g., 512 bytes.
- Mostly used in combination with a file system (such as ext4 or similar).

Blob/Object Storage

- Access always takes place via REST API.
- Objects (binary files, structured text files -> files) are written en bloc.
- Specific extensions are append blobs.

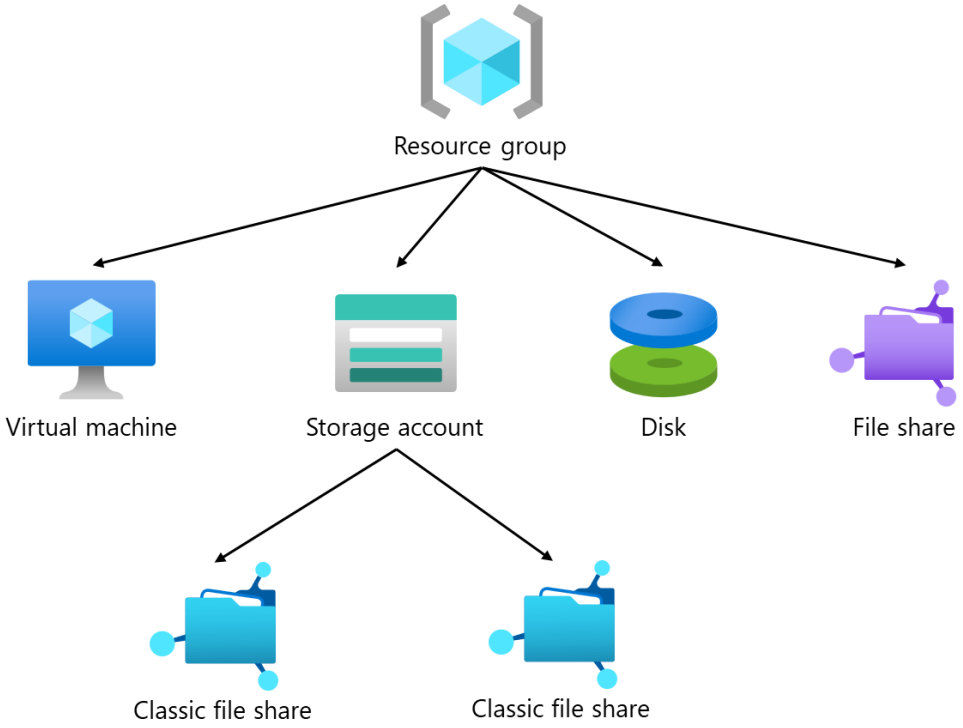
File Storage

- NFS/SMB-compatible storage solution
- Supports various storage types such as SSD or HDD
- Mounting via PowerShell or Bash script
- Use cases
  - POSIX-compatible file system required
  - Multiple clients must be able to work on the same file system
  - Shared file storage



File Storage

- Transition process from “storage account file shares” to “file shares” (service renaming)
- Decoupling of storage accounts
- File shares are still in preview
- Additional premium offerings: Netapp Files




```
az storage share create --account-name MyAccount --name MyFileShare
```

https://learn.microsoft.com/en-us/azure/storage/files/storage-files-planning

File Storage - Pricing

- Depending on SKU, redundancy, region, storage amount, and IOPS
- Additionally, depending on usage, snapshots and soft delete

Redundancy: LRS | Region: Switzerland North | Currency: Switzerland – Franc (chf) CHF | Display pricing by: Month

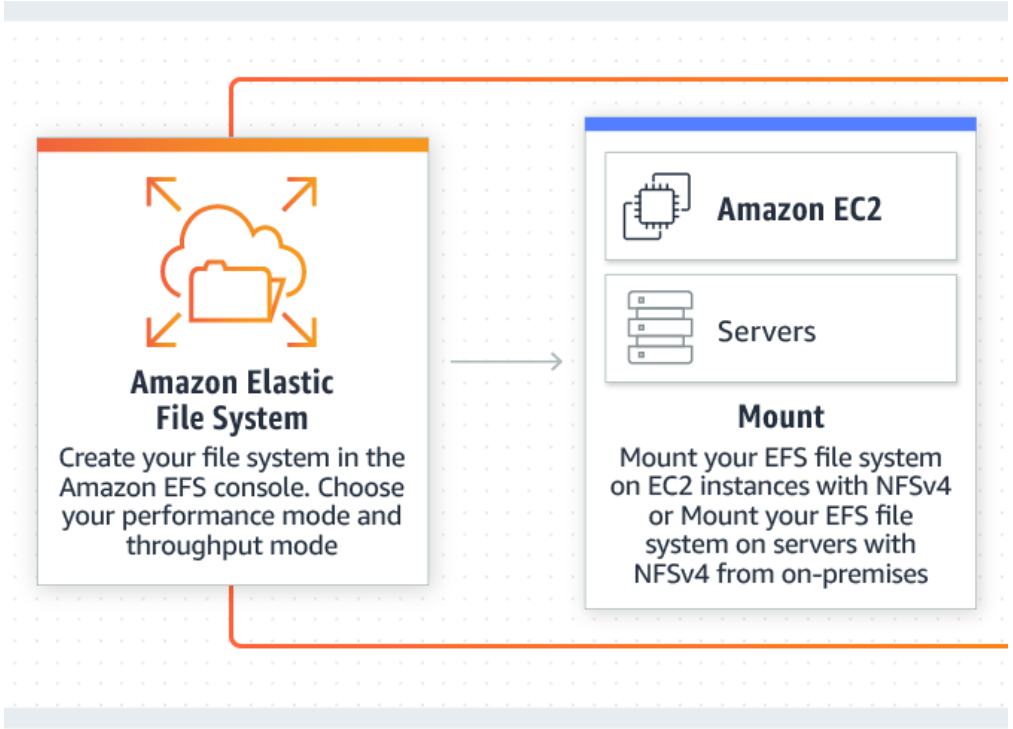
1 USD = 0.804 CHF

Quantity	SSD Price	HDD Price
Provisioned storage The maximum storage size of the file share, specified in GiB. You are charged for the storage you've provision regardless of what you use. Provisioned storage can be changed after share creation.	CHF 0.1151 per provisioned GiB per month	CHF 0.0083 per provisioned GiB per month
Provisioned IOPS The maximum IOPS, or transactions per second, of the file share. You are charged for the IOPS you've provisioned regardless of how many IOs you use. Provisioned IOPS can be changed after share creation.	CHF 0.0312 per provisioned IO per sec per month *First 3000 IOPS at no additional cost*	CHF 0.0464 per provisioned IO per sec per month
Provisioned throughput The maximum amount of data that can be transferred per second of the file share. You are charged for the throughput you've provisioned regardless of what you use. Provisioned throughput can be changed after share creation.	CHF 0.0452 per provisioned MiB per sec per month *First 100 MiB / sec at no additional cost*	CHF 0.0687 per provisioned MiB per sec per month

https://azure.microsoft.com/de-de/pricing/details/storage/files/

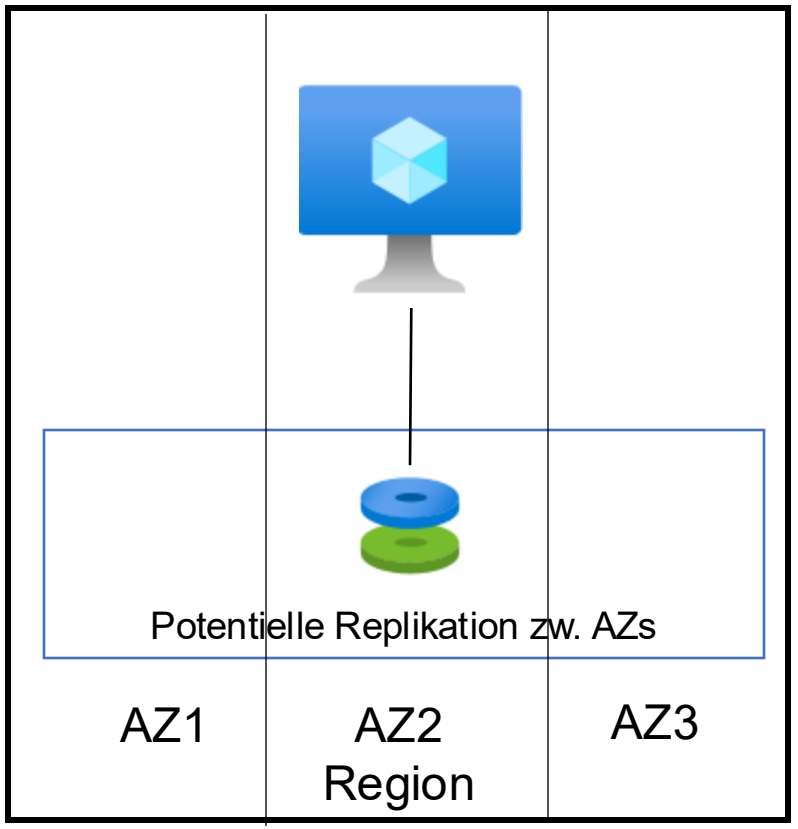
File Storage – AWS Elastic File System (EFS)

- File storage product from AWS, comparable to Azure Files
- Available as NFS, SMB, or Linux storage
- Supports various storage classes (SSD, hybrid, SATA disk)
- Additional specialized services: Amazon FSx for NetApp ONTAP or Amazon FSx for Windows File Server



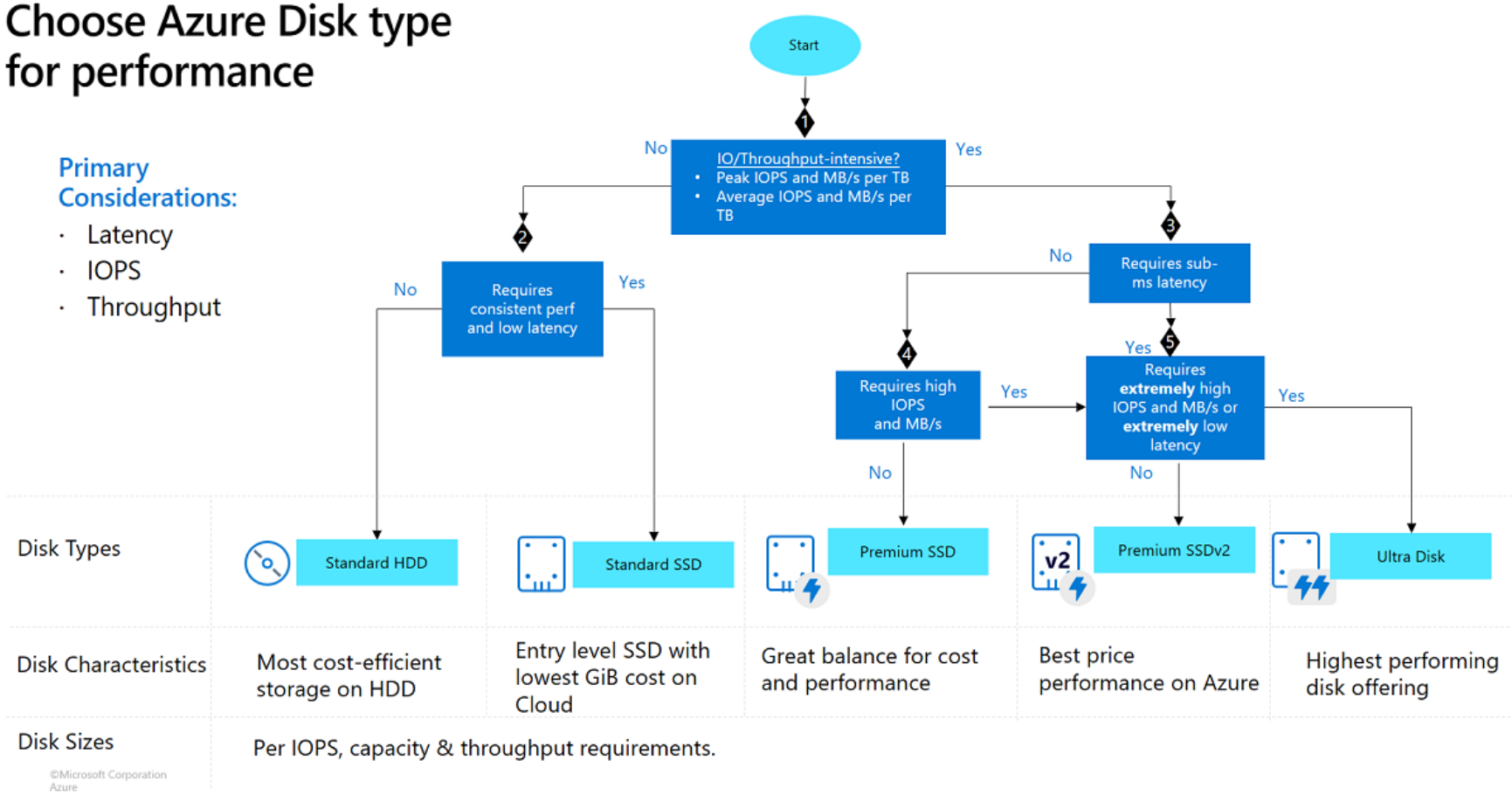
Block Storage

- Native storage solution for virtual machines in the form of disks
- Performance classes such as SSD, HDD, or Ultra
- Features
  - Snapshots
  - Replication (LRS, ZRS)
  - Encryption
  - Base images



Block Storage

- Selecting the disk type based on performance requirements
- Note: Reserved capacity and IOPS have a significant impact on costs



https://learn.microsoft.com/en-us/azure/virtual-machines/disks-types

Azure Managed Disk

- Azure Disks have a lifecycle that is independent of virtual machines
- Azure Disks can be created from
  - Scratch (empty disk)
  - Snapshots of other disks
  - Base images (e.g., Ubuntu)

```
az disk create -g MyResourceGroup -n MyDisk --size-gb 10
```

```
az vm disk attach \
   -g myResourceGroup \
   --vm-name myVM \
   --name myDataDisk \
  [--new \]
   --size-gb 50
```

https://learn.microsoft.com/en-us/cli/azure/disk?view=azure-cli-latest#az-disk-create

Azure Managed Disk

- Snapshots are point-in-time backups of disks.
- Snapshots can be created incrementally.
- Snapshots have their own lifecycle and are not dependent on the source (e.g., a disk).
- Billing is based on the storage space actually used.

Create a snapshot:

```
az snapshot create -g MyResourceGroup -n MySnapshot2 --source MyDisk
```

Create a disk from a snapshot:

```
az disk create -g MyResourceGroup -n MyDisk2 --source MyDisk
```

Snapshots are crash consistent!

https://learn.microsoft.com/en-us/cli/azure/snapshot?view=azure-cli-latest#az-snapshot-create
https://learn.microsoft.com/en-us/azure/virtual-machines/managed-disks-overview#snapshots

Azure Managed Disk

- Azure Storage Services (whether disk, blob, DBs, etc.) are always encrypted at rest.
- The encryption key is platform-managed by default.
- Advanced encryption options with Customer Managed Key (CMK) are supported.

Storage Server Side Encryption
Standard encryption method from storage. Can be configured with CMK.

Encryption at Host
Encryption directly on the host before the blocks are transferred to storage (end-to-end).

Azure Disk Encryption
Encryption of the OS and data directly on the disk (via BitLocker / DM-Crypt).

Confidential Disk Encryption
Encryption using TPM directly in the VM. OS-only supported.

https://learn.microsoft.com/en-us/azure/storage/common/storage-service-encryption
https://learn.microsoft.com/en-us/azure/virtual-machines/disk-encryption-overview

Block Storage – AWS Elastic Block Store (EBS)

- Equivalent of AWS to Azure Disks
- AWS EBS Volumes = Azure Disks
- EBS Volume types comparable to Azure Disk SKUs

AWS EBS volume type	Azure Managed disk	Use	Can this managed disk be used as an OS Disk
gp2/gp3	Standard SSD	Web servers and lightly used application servers or dev/test environments	Yes
gp2	Premium SSD	Production and performance-sensitive workloads	Yes
gp3	Premium SSD v2	Performance-sensitive workloads or workloads that require high IOPS and low latency	No
io2	Ultra Disk Storage	IO-intensive workloads, performance-demanding databases, and very high transaction workloads that demand high throughput and IOPS	No

On both platforms: The VM SKU must match the block storage SKU. For example, small VMs cannot take advantage of the performance of Ultra Disks!

https://learn.microsoft.com/en-us/azure/architecture/aws-professional/storage

Storage Products – Blob / Object Storage

AWS S3

- Buckets as data containers for objects
- Buckets are regional and can be replicated
- Access via globally unique URL

Azure Blob Storage

- Storage Account -> Container -> Blobs
- Storage accounts specify the region (can be replicated)
- Access via globally unique URL

GCP Cloud Storage

- Buckets as data containers for objects
- Buckets are regional and can be replicated
- Access via globally unique bucket names. Base URL is stable.

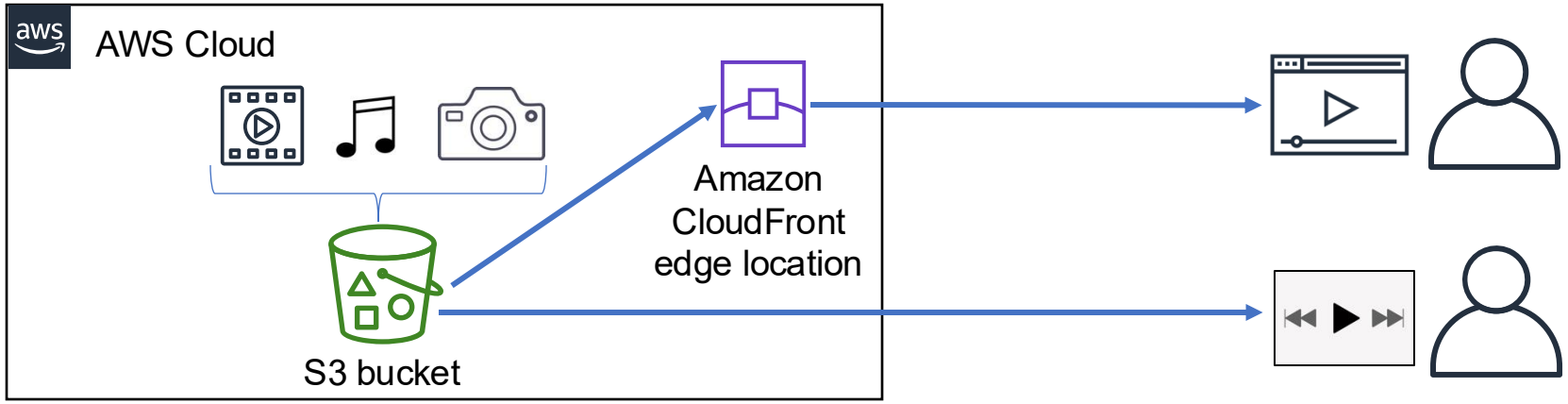
There are similar features, but the APIs are not compatible!

Object / Blob Storage – Use Cases

A redundant, scalable, and highly available infrastructure that enables the uploading and downloading of videos, photos, or music.

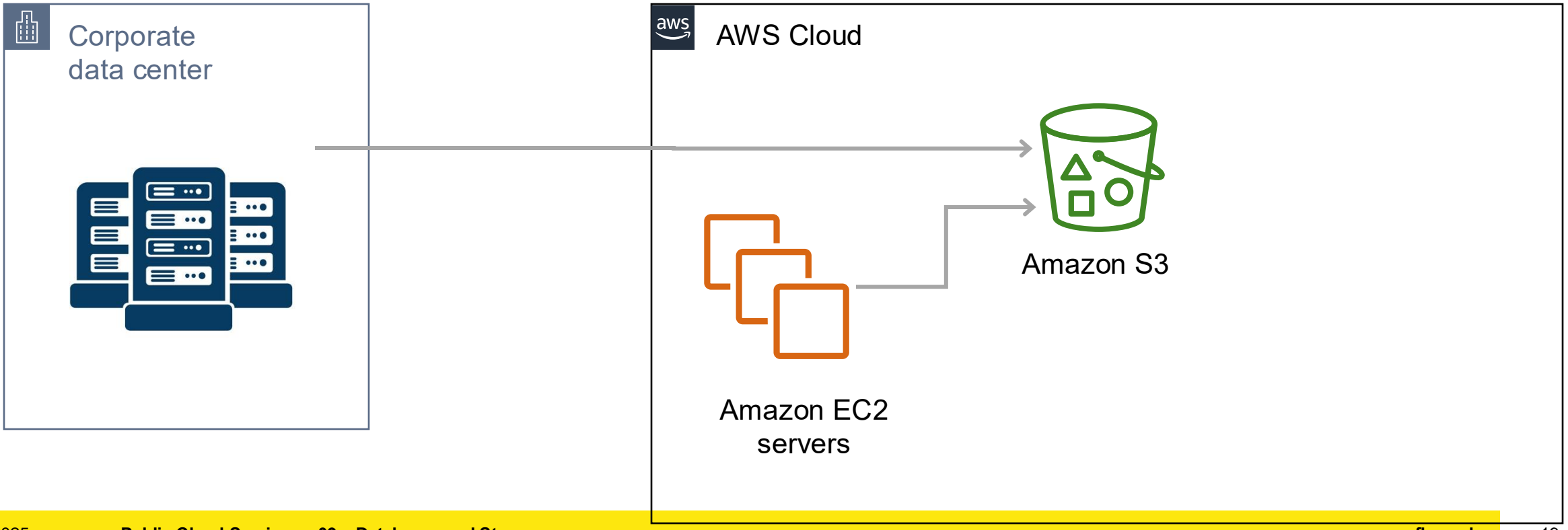
`https://<bucket-name>.s3.amazonaws.com`

`https://<bucket-name>.s3.amazonaws.com/video.mp4`



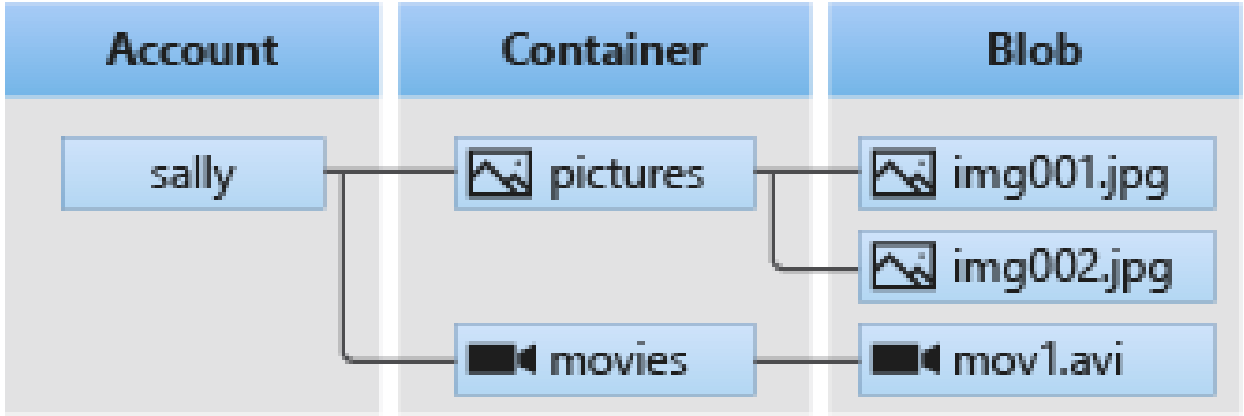
Object / Blob Storage – Use Cases

Object Storage as backup



Azure Storage Account

- Storage accounts are logical containers for multiple Azure storage services (blob, file, table, and queue).
- A storage account can contain N blob containers (similar to S3 buckets).
- The structure within the container is similar to folder structures (folders and blobs).

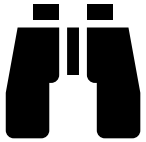


https://learn.microsoft.com/en-us/azure/storage/blobs/storage-blobs-introduction

Azure Storage Account

Durability

- Storage account durability between 11 nines and 16 nines, depending on SKU



Availability

- 99.9% availability SLA for hot tier (varies depending on SKU)

Scalability

- ~5 PB per Storage Account
- ~190 TB per Blob

Security

- Access via EntraID or access keys

Performance

- Support for various access patterns (block blob, page blob, append blob)

https://learn.microsoft.com/en-us/azure/storage/common/scalability-targets-standard-account

Azure Storage Account

- Storage account names must be globally unique.
- Containers must be unique at the same level in the storage account.

When naming your storage account, keep these rules in mind:

- Storage account names must be between 3 and 24 characters in length and may contain numbers and lowercase letters only.
- Your storage account name must be unique within Azure. No two storage accounts can have the same name.

https://pclsdemo.blob.core.windows.net/assets/static/ag.svg

Globally unique name

Container name
(2 Container)

Blob name

https://learn.microsoft.com/en-us/azure/storage/common/storage-account-overview

Azure Storage Account - Redundancy

Storage accounts are region-level resources. The region determines the data residency.

Redundancy	Description	Benefits
LRS Locally Redundant Storage	Redundant within the region. No control over AZ in the region.	Cost-effective 3 replicas of the data Writes are synchronous
ZRS Zone Redundant Storage	Replication of data across 3 or more zones in the region.	Fail-safe for AZ failures Writes are synchronous
GRS / Geo-Redundant Storage GZRS / Geo-Zone Redundant Storage	Replication of data to region pair. Distribution analogous to LRS/ZRS.	Fail-safe for regions Writes synchronously in primary region and asynchronously in secondary region

https://learn.microsoft.com/en-us/azure/storage/common/storage-redundancy

Azure Storage Account - Concurrency

- Azure Storage Account supports three concurrency models
- Standard: Last Writer Wins

Optimistic Concurrency

- The application must check whether the data has been changed during write operations.
- Applications ensure that write operations do not overwrite each other.
- Error handling for the user is required.

Pessimistic Concurrency

- The application creates a lock on the object to be modified
- Error handling may be necessary for objects with locks

Last Writer Wins

- Die Applikation schreibt direkt, ohne zu prüfen ob die Daten in der Zwischenzeit geändert wurden

https://learn.microsoft.com/en-us/azure/storage/blobs/concurrency-manage#optimistic-concurrency

Azure Storage Account – Lifecycle Policies

- Automated management of objects with increasing storage requirements
- Use cases
  - Archiving data after, for example, 90 days without access
  - Deleting data after, for example, 180 days
  - Deleting “old” versions of data, but retaining the “latest” version
- Example policy Deletes blobs after 365 days without modification

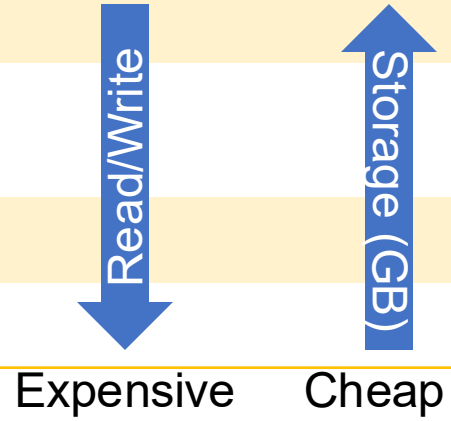
```
{
  "rules": [
    {
      "name": "expirationRule",
      "enabled": true,
      "type": "Lifecycle",
      "definition": {
        "filters": {
          "blobTypes": [ "blockBlob" ]
        },
        "actions": {
          "baseBlob": {
            "delete": { "daysAfterModificationGreaterThan": 365 }
          }
        }
      }
    }
  ]
}
```

https://learn.microsoft.com/en-us/azure/storage/blobs/lifecycle-management-policy-delete

Azure Storage Account – Access Tiering

Access tiers serve to optimize costs, access speed, and retention depending on the use case.

Tier	Description	Cost Structure	Access Speed
Hot	High-frequency use of data		Immediately
Cool	Low-frequency use of data		Immediately
Cold	Rare use of data		Immediately
Archive	Archiving of data		Up to 15h to rehydrate



https://learn.microsoft.com/en-us/azure/storage/blobs/storage-blobs-introduction

Azure Storage Account - Emulator

- Storage Account equivalent solution for local development / offline development
- Storage Explorer: Legacy solution
- Azurite Emulator: Successor to the Storage Emulator, feature set not yet on par

https://learn.microsoft.com/en-us/azure/storage/common/storage-use-azurite

Azure Storage Account - Permissions

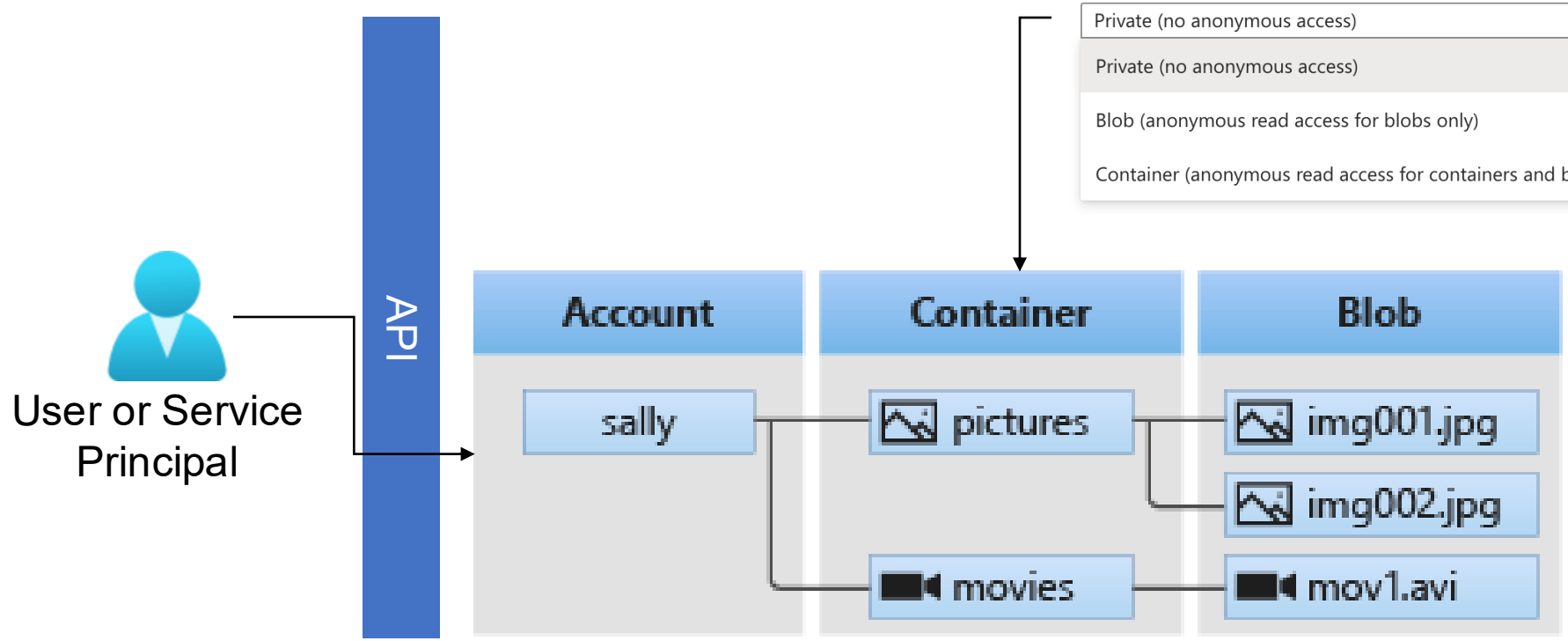
- Permission options vary depending on storage type (blob, file, queue, etc.)
- Distinction between access via
  - EntraID and Azure RBAC
  - Storage account access keys (or SAS keys derived from them)
  - Anonymous

Authorization option	Guidance	Recommendation
Microsoft Entra ID	Authorize access to Azure Storage data with Microsoft Entra ID	Microsoft recommends using Microsoft Entra ID with managed identities to authorize requests to blob resources.
Shared Key (storage account key)	Authorize with Shared Key	Microsoft recommends that you disallow Shared Key authorization for your storage accounts.
Shared access signature (SAS)	Using shared access signatures (SAS)	When SAS authorization is necessary, Microsoft recommends using user delegation SAS for limited delegated access to blob resources. SAS authorization is supported for Blob Storage and Data Lake Storage, and can be used for calls to `blob` endpoints and `dfs` endpoints.
Anonymous read access	Overview: Remediating anonymous read access for blob data	Microsoft recommends that you disable anonymous access for all of your storage accounts.
Storage Local Users	Supported for SFTP only. To learn more, see Authorize access to Blob Storage for an SFTP client	See guidance for options.
Microsoft Purview protection policies	Supported for Azure Blob Storage and Azure Data Lake Storage. To learn more, see Authoring and publishing protection policies for Azure sources (Preview).	See guidance for options.

https://learn.microsoft.com/en-us/azure/storage/common/authorize-data-access?tabs=blobs

Azure Storage Account – Permissions for Blobs – Blob Access Level

- Access levels are defined per (top-level) container
- Anonymous access can be disabled for storage accounts

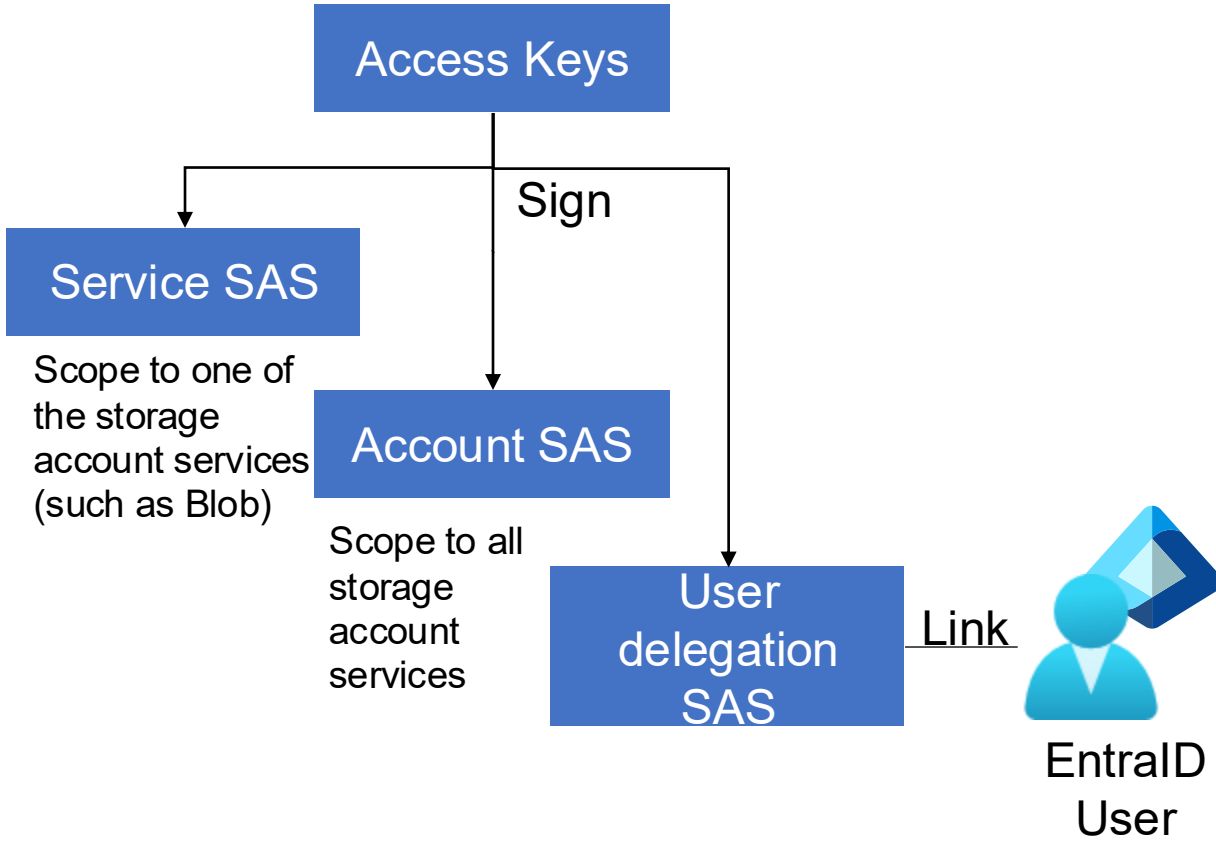



```
curl https://pclsdemo.blob.core.windows.net/assets/static/ag.svg > ag.svg
```

https://learn.microsoft.com/en-us/azure/storage/common/authorize-data-access?tabs=blobs

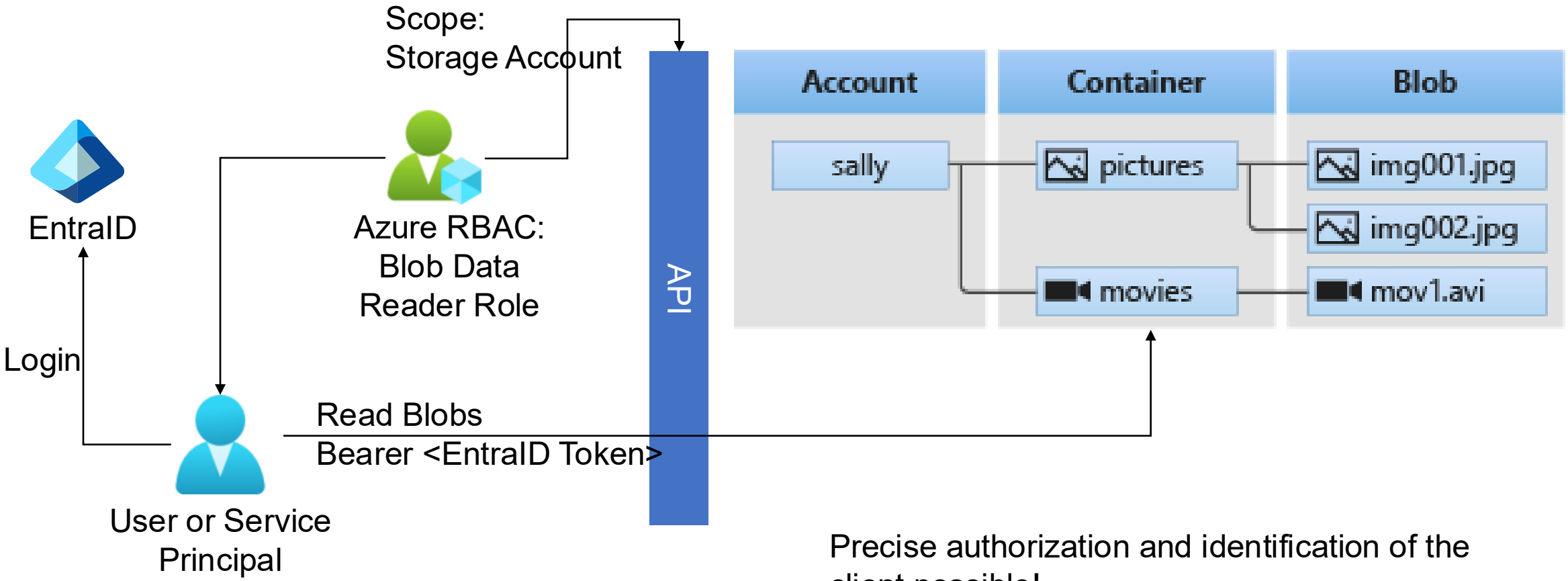
Azure Storage Account – Permissions for Blobs – Access Keys

- Access keys are administrative keys for all data and management operations.
- Access keys are used to sign SAS keys (shared access signature).
- The user/service is usually not identifiable.
- Access keys can be deactivated.
- SAS are used by clients to access data in blob storage.



https://learn.microsoft.com/en-us/azure/storage/common/authorize-data-access?tabs=blobs

Azure Storage Account – Permissions for Blobs - EntraID



Precise authorization and identification of the client possible!

https://learn.microsoft.com/en-us/azure/storage/common/authorize-data-access?tabs=blobs

Azure Storage Account – Dataimport

- Importing large amounts of data into Azure Storage Accounts / AWS S3
- Data volumes are in the petabyte (Snowball) range, starting at 40 terabytes with DataBox
- Transfer via the Internet / VPN not possible due to volume or bandwidth

Azure DataBox

Azure Import/Export

AWS Snowball

AWS Snowmobile

https://learn.microsoft.com/de-de/azure/databox/data-box-overview?pivots=dbx-ng /
https://aws.amazon.com/de/snowball/

Azure Storage Account – Dataimport

- Data import via network (VPN or Internet) into Azure Storage Accounts is performed using
  - AzCopy (CLI)
  - Azure Data Factory (data integration solution)
  - Blobfuse (POSIX wrapper for Blob API)
  - Azure Storage Data Movement Library for .NET
- Sources can be AWS S3 or existing file shares, for example.

Single files:

```
azcopy copy '<local-file-path>' 'https://<storage-account-name>.<blob or dfs>.core.windows.net/<container-name>/<blob-name>'
```

Folders:

```
azcopy copy '<local-directory-path>' 'https://<storage-account-name>.<blob or dfs>.core.windows.net/<container-name>' --recursive
```

https://learn.microsoft.com/en-us/azure/storage/blobs/storage-blobs-introduction#move-data-to-blob-storage

Azure Storage Account – Pricing

- Cost factors for blob storage
  - Data volume
  - Access frequency
  - Access tier
- Optimization through
  - Choice of the appropriate access ti
  - Use of lifecycle policies
  - Reserved capacity

Data storage prices pay-as-you-go	Premium	Hot	Cool	Archive
First 50 terabyte (TB)/month	CHF 0.17968 per GB	CHF 0.0176 per GB	CHF 0.00879 per GB	CHF 0.00159 per GB
Next 450 TB/month	CHF 0.17968 per GB	CHF 0.0169 per GB	CHF 0.00879 per GB	CHF 0.00159 per GB
Over 500 TB/month	CHF 0.17968 per GB	CHF 0.0162 per GB	CHF 0.00879 per GB	CHF 0.00159 per GB

Operations and data transfer	Premium	Hot	Cool	Archive
Write operations (per 10,000)[1]	CHF 0.0274	CHF 0.0561	CHF 0.1039	CHF 0.1371
Read operations (per 10,000)[1]	CHF 0.0022	CHF 0.0045	CHF 0.0104	CHF 6.8516
Archive High Priority Read (per 10,000)[1]				CHF 74.2253
Iterative Read Operations (per 10,000)[1]	CHF 0.0274	CHF 0.0045	CHF 0.0045	CHF 0.0045

https://azure.microsoft.com/en-us/pricing/details/storage/blobs/

Storage – Summary

- Block storage
  - Directly attached to VMs in the form of disks
  - Use only via VMs
- File storage
  - POSIX-compatible storage in the form of shares
  - Use by VMs and other services via network
- Blob storage
  - Object storage in the form of buckets or storage account containers
  - Use by any clients via REST API

Demo

- Create storage account with IaC
- Upload/download blob via keys

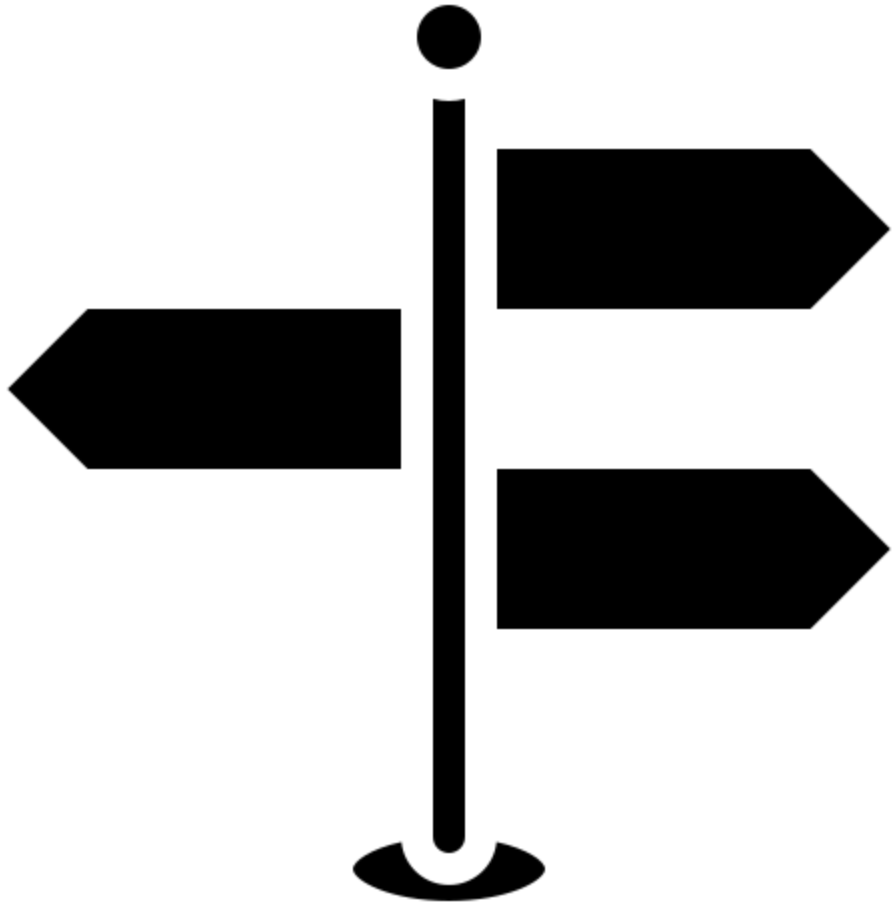
Exercise

- Create your own storage account via the portal.
- Make blobs accessible on a time-based basis via shared access signature (i.e. access expiration after a few mins)
- Make blobs publicly accessible with anonymous access.

https://learn.microsoft.com/en-us/azure/storage/blobs/storage-blobs-introduction

Learning Objectives

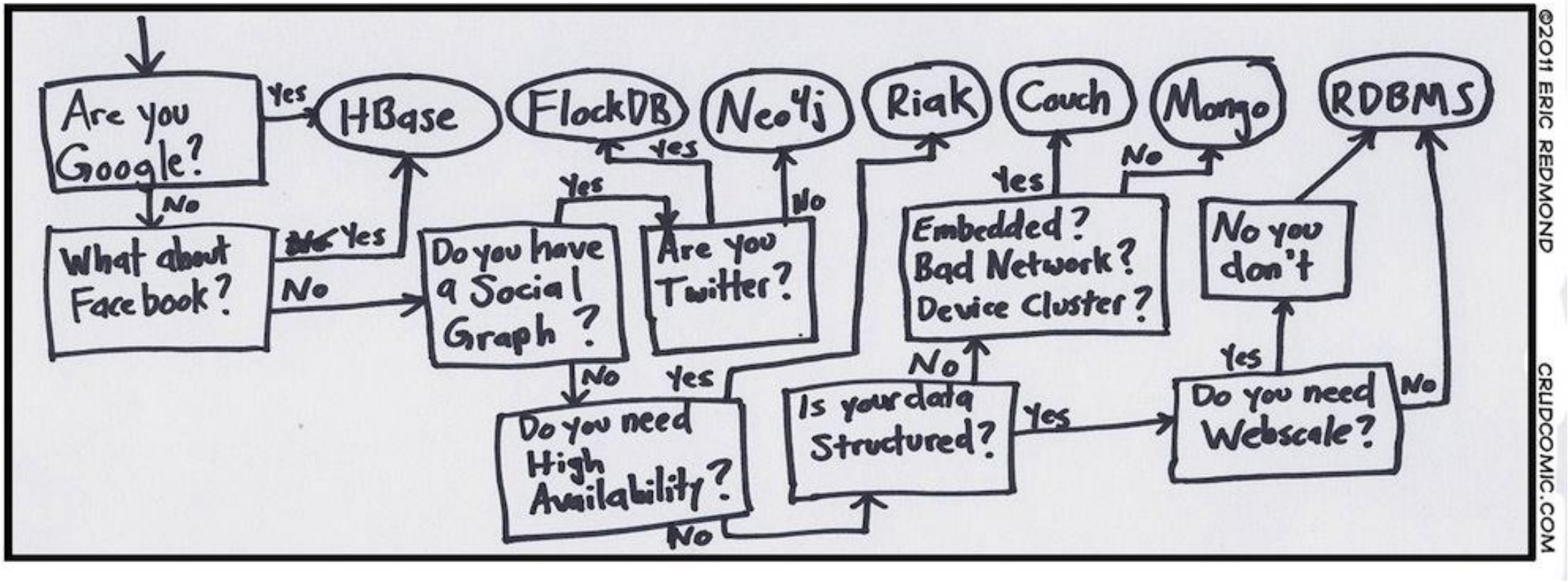
- Students gain an overview of DBaaS (Database as a Service) and understand the challenges of operating databases in the cloud.
- Students can differentiate between different types of DBaaS and use them correctly in context.
- Students understand DBaaS licensing models.



Agenda

- CAP
- DB Services
- Example: Cosmos and Postgres
- Integration into app
- IaC

Motivation


Are you Google?
Yes
HBase
No
What about Facebook?
Yes
No
Do you have a Social Graph?
Yes
No
FlockDB
Yes
Neo4j
No
Are you Twitter?
Yes
No
Do you need High Availability?
Yes
No
Riak
Couch
Yes
Embedded? Bad Network? Device Cluster?
No
Mongo
No
No you don't
RDBMS
Is your data Structured?
Yes
Yes
Do you need Webscale?
No
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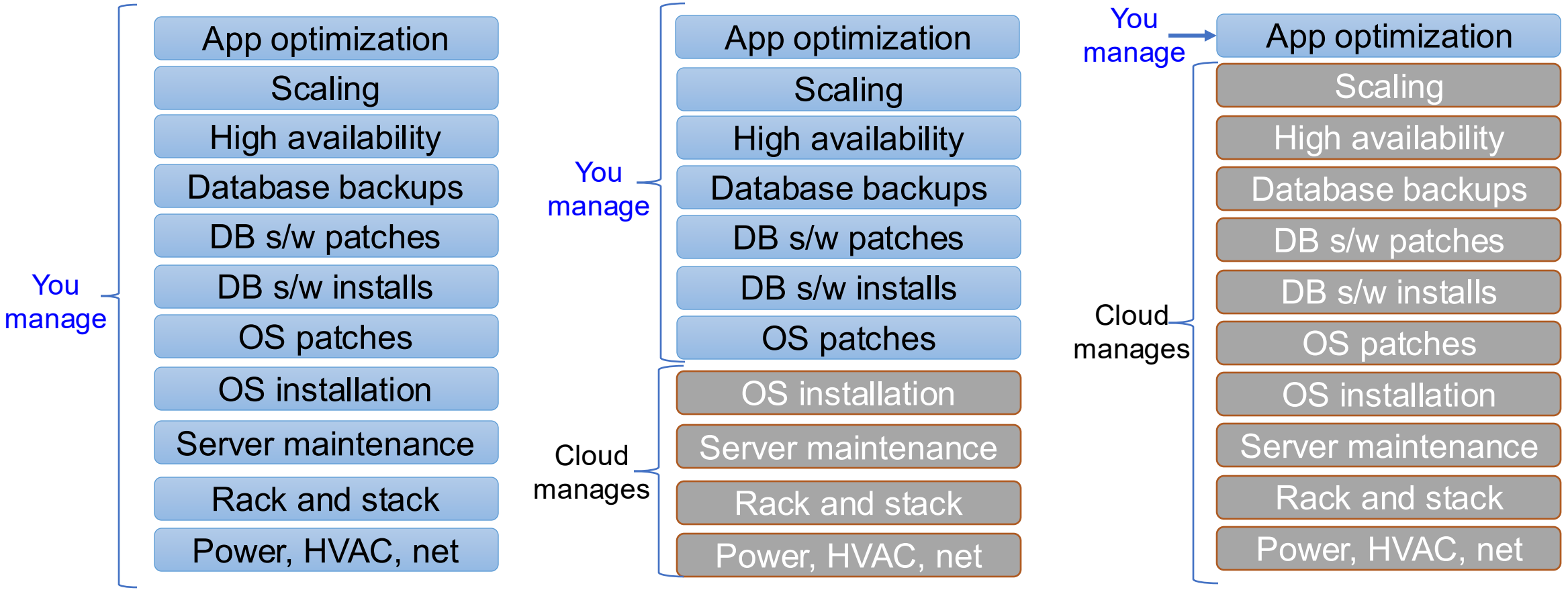
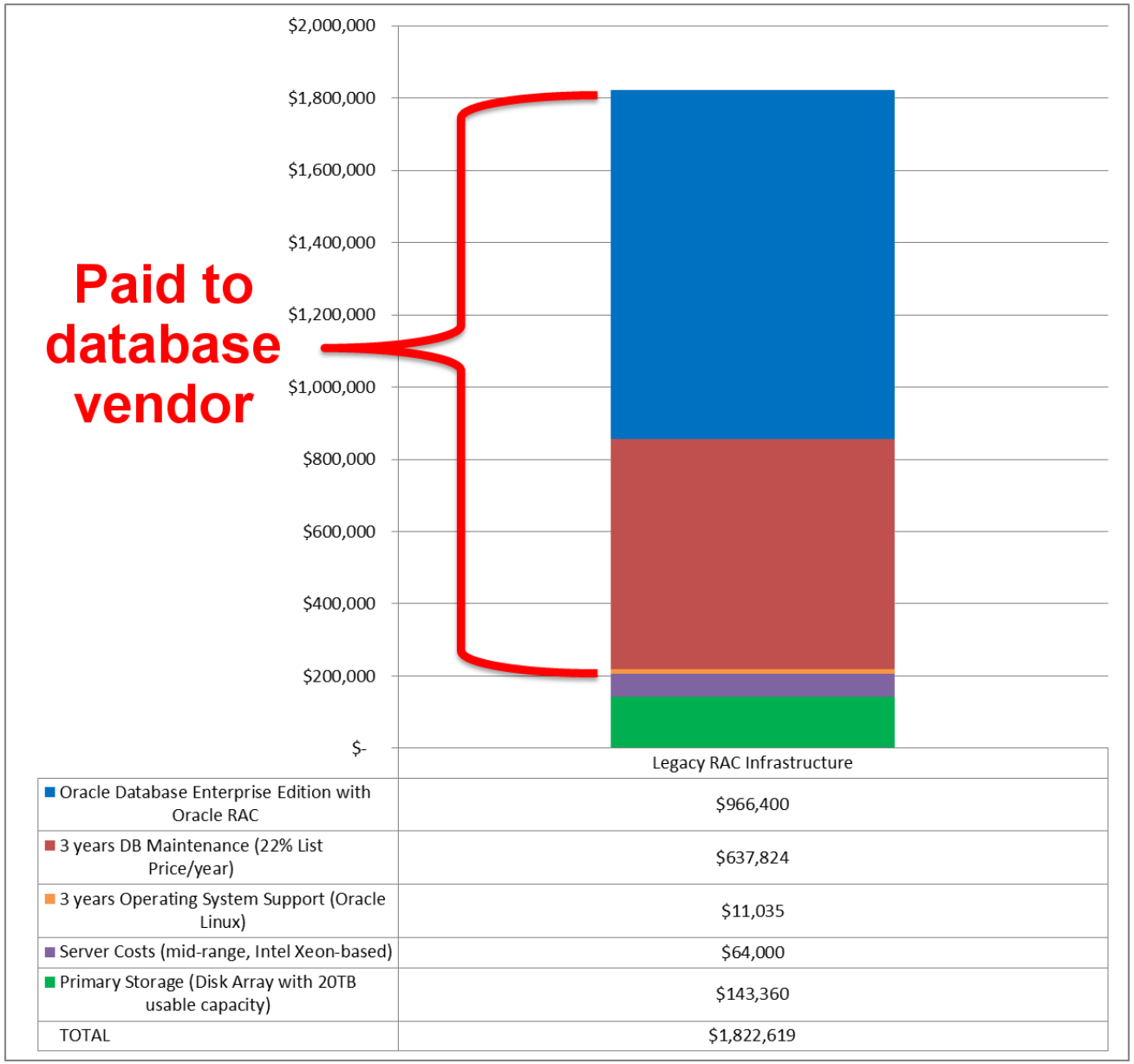
Why Cloud Databases



Why Cloud Databases



	Legacy RAC Infrastructure
Oracle Database Enterprise Edition with Oracle RAC	$966,400
3 years DB Maintenance (22% List Price/year)	$637,824
3 years Operating System Support (Oracle Linux)	$11,035
Server Costs (mid-range, Intel Xeon-based)	$64,000
Primary Storage (Disk Array with 20TB usable capacity)	$143,360
TOTAL	$1,822,619

CAP-Theorem

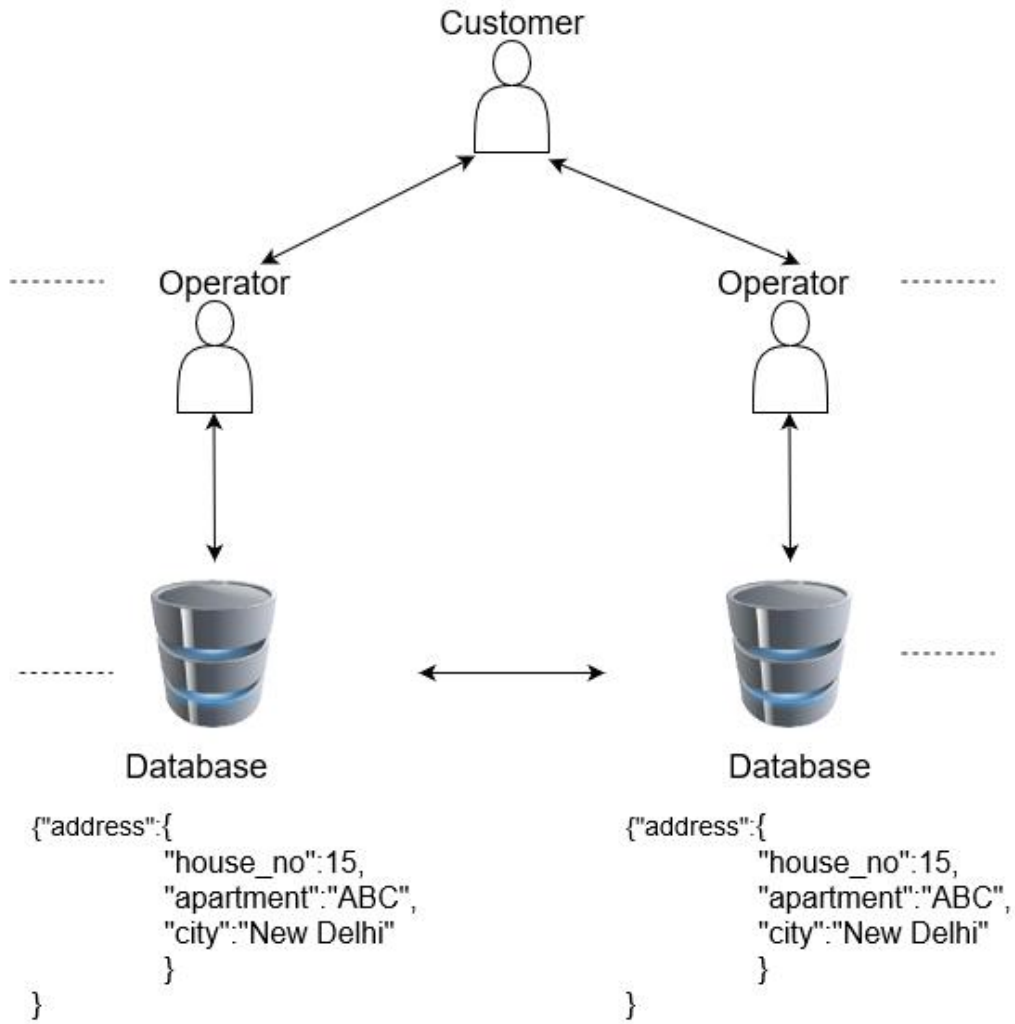
"It is impossible for a web service to provide these *three guarantees at the same time (pick 2 of 3)*:

- **(Sequential) Consistency**
- **Availability**
- **Partition-tolerance**"

Eric Brewer: https://en.wikipedia.org/wiki/CAP_theorem

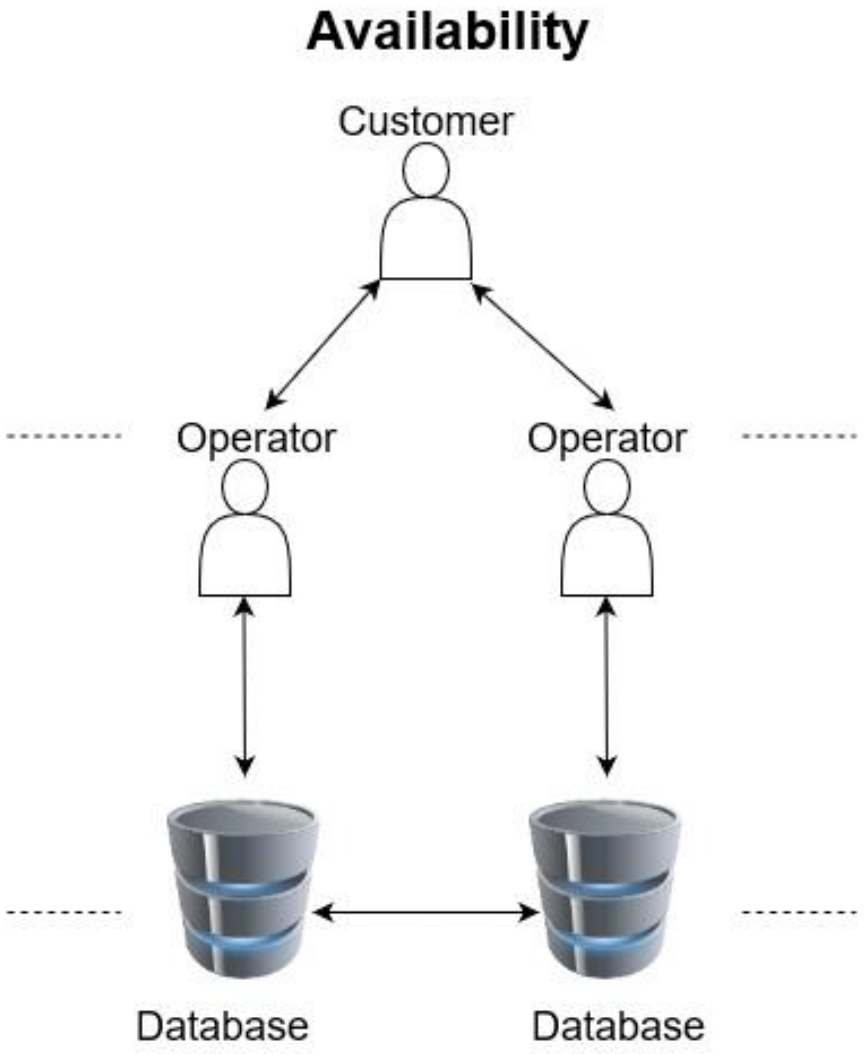
Consistency

- All nodes return the same data at the same time
- For successful write operations, all read operations subsequently respond to the new state



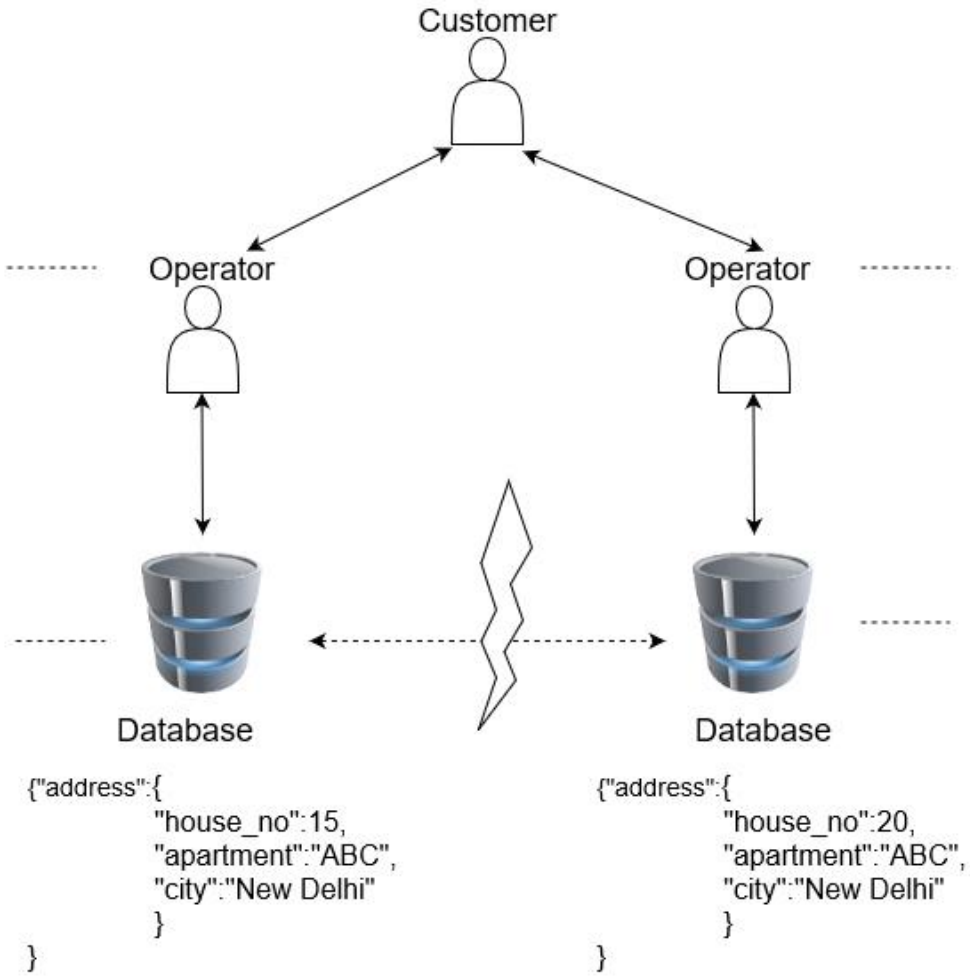
Availability

- Each functioning node can respond to requests without waiting for other nodes...
- also applies to write accesses
- Requests may receive outdated data as a response



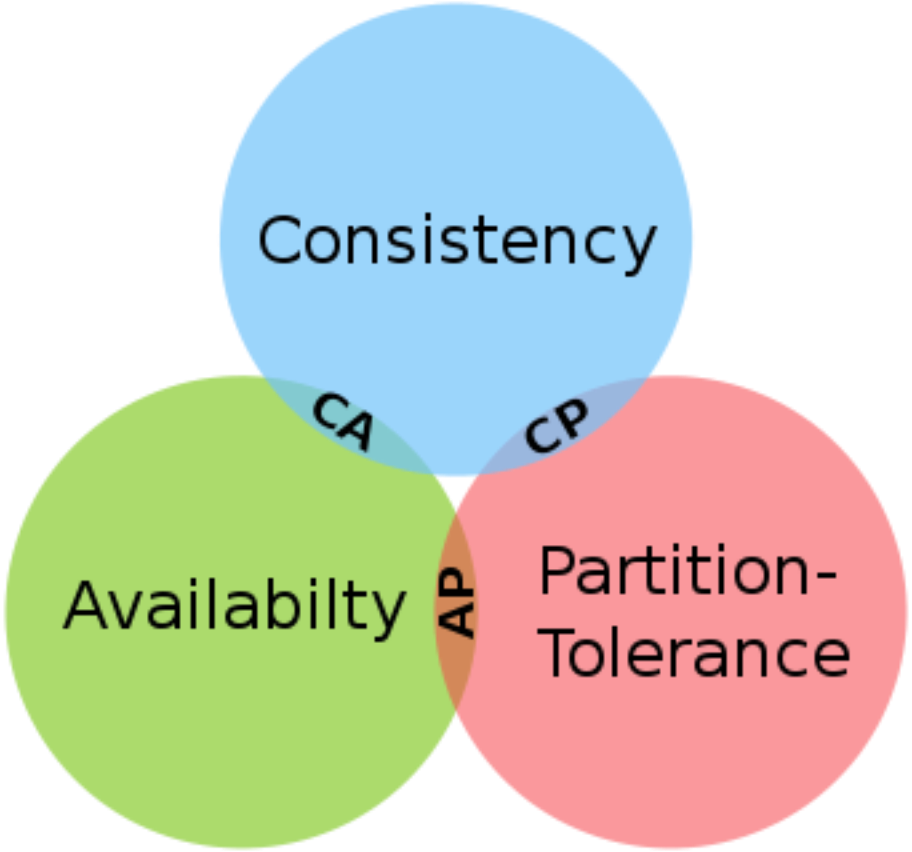
Partition Tolerance

- If the nodes can no longer communicate with each other, the individual nodes must still function normally on their own.
- P is unavoidable in distributed systems:
  - “Of the CAP theorem's Consistency, Availability, and Partition Tolerance, Partition Tolerance is mandatory in distributed systems. You cannot not choose it.” – Coda Hale
  - “An important observation is that in larger distributed-scale systems, network partitions are a given; therefore, consistency and availability cannot be achieved at the same time” – Werner Vogels



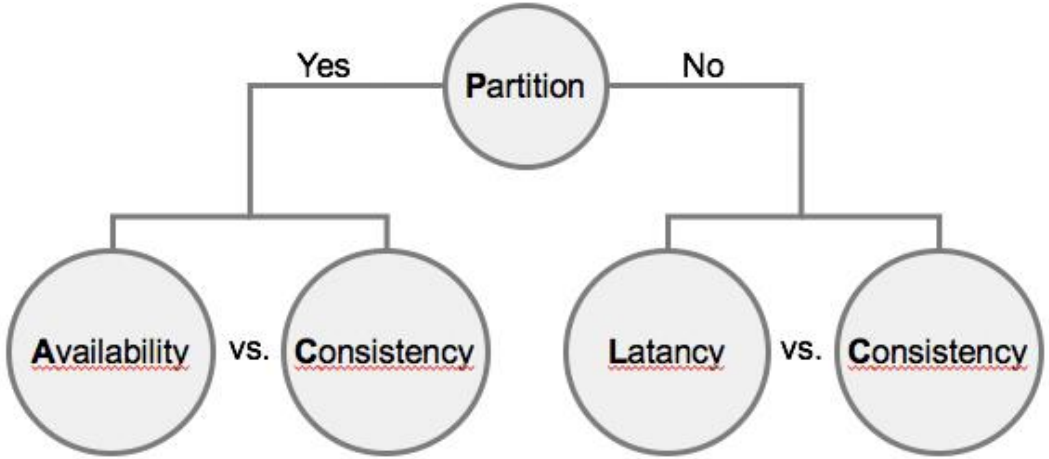
Implications on distributed systems

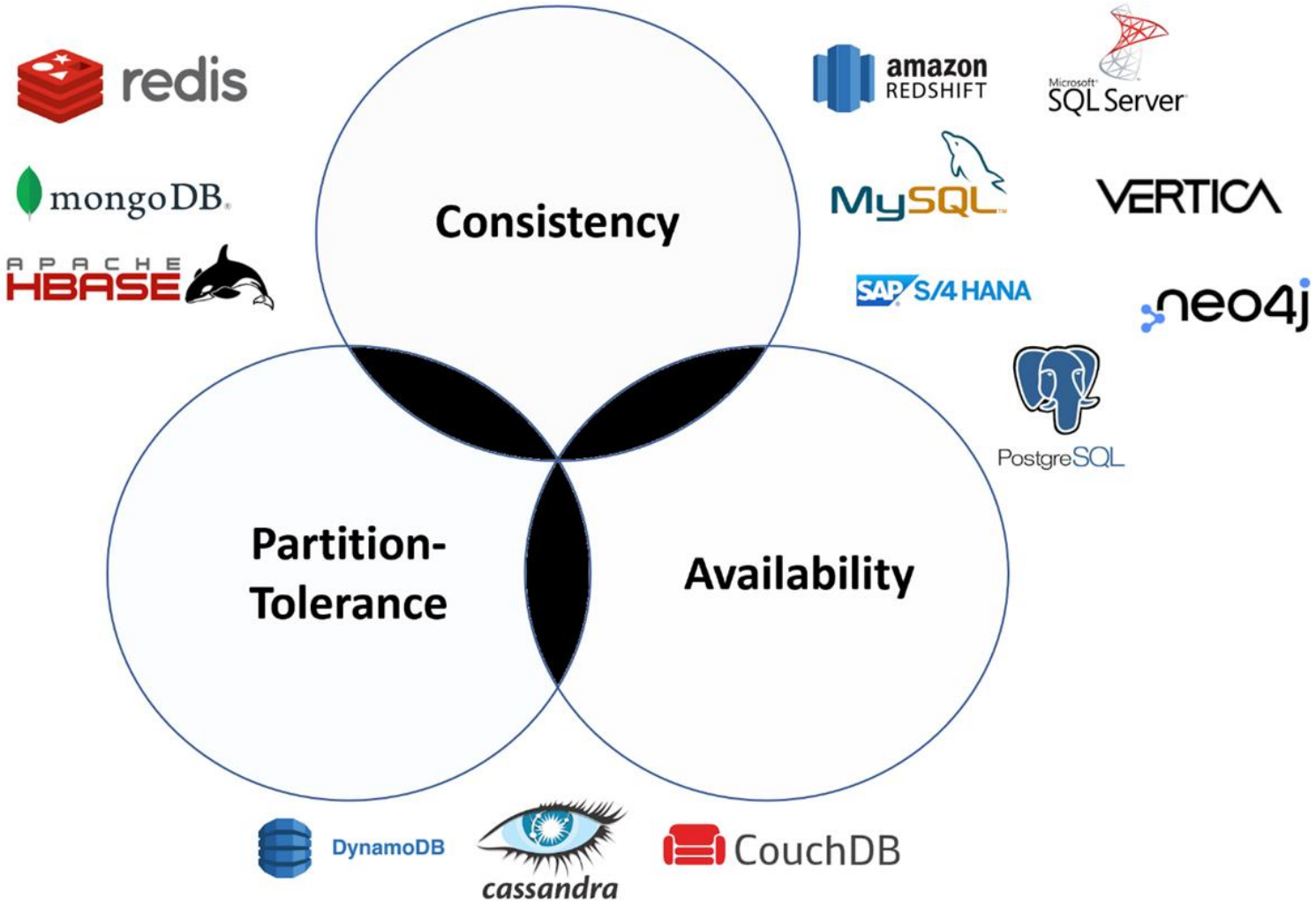
- Of the following guarantees in distributed systems:
  - Consistency
  - Availability
  - Partition tolerance
- You can choose a maximum of two at the same time.
- This results in three types of distributed systems for storing data:
  - CP
  - AP
  - CA



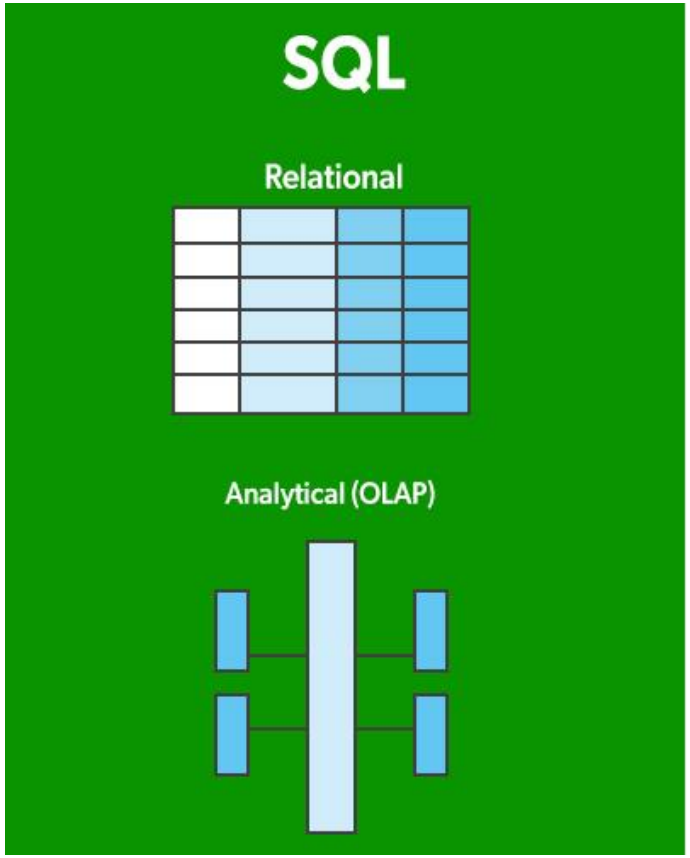
Summary CAP

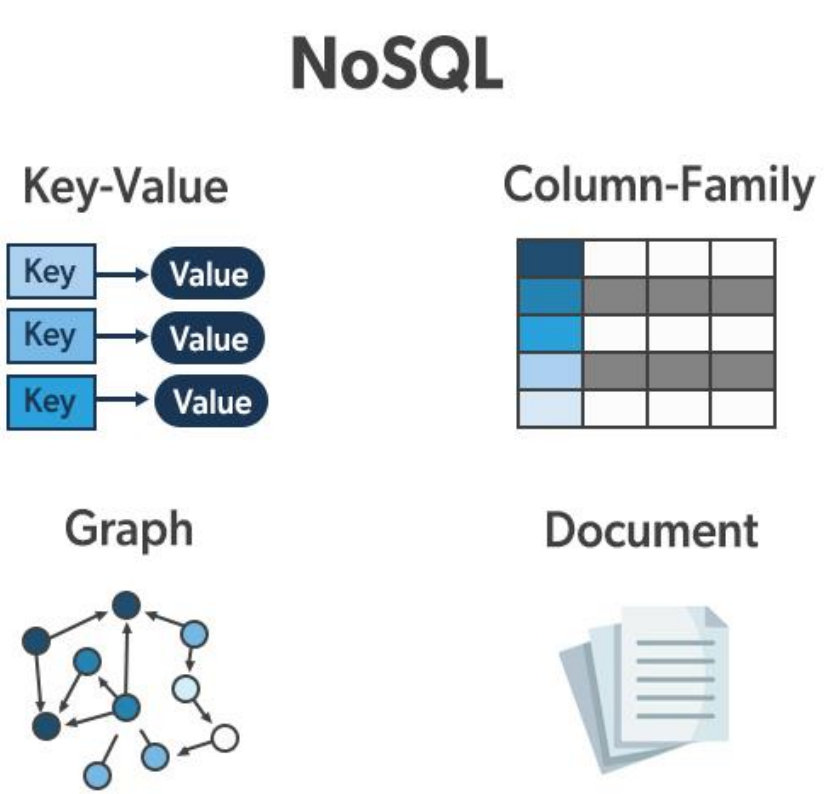
- CAP should stimulate thinking: What is important in a "distributed" architecture?
  - The devil is in the details: Complete architectures cannot be "simply" classified into CA/AP/CP.
  - Databases, however, tend to be...
- Above all, it should be clear how a system behaves in a network partitioning case.
  - If there is no partitioning, everything should be consistent and available anyway.
  - Many data stores can still fall back on CAP.
- There is another extension: PACELC.



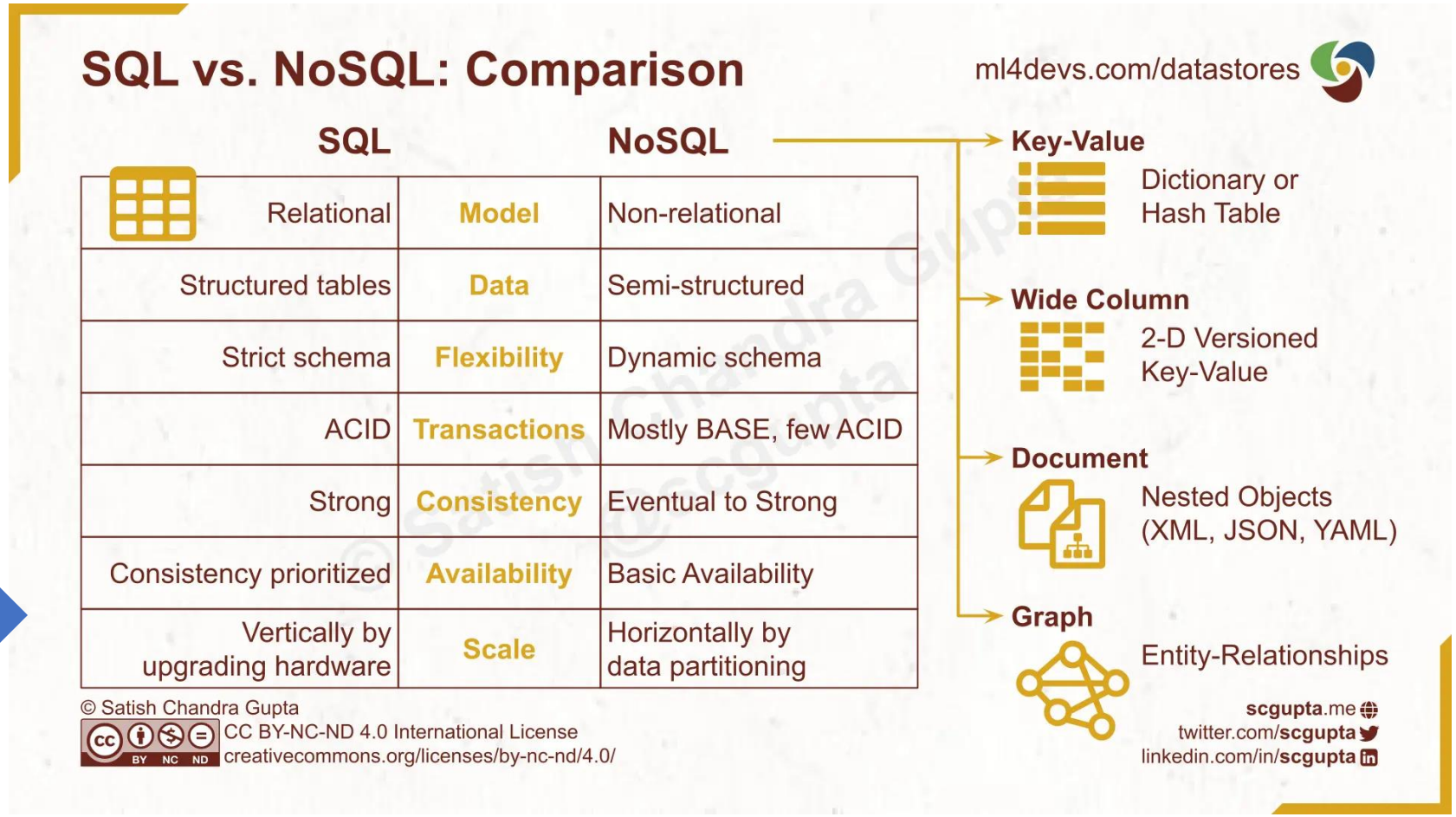

redis
mongoDB.
APACHE HBASE
amazon REDSHIFT
Microsoft SQL Server
MySQL
VERTICA
SAP S/4 HANA
neo4j
PostgreSQL
Consistency
Partition-Tolerance
Availability
DynamoDB
cassandra
CouchDB

Database Classification



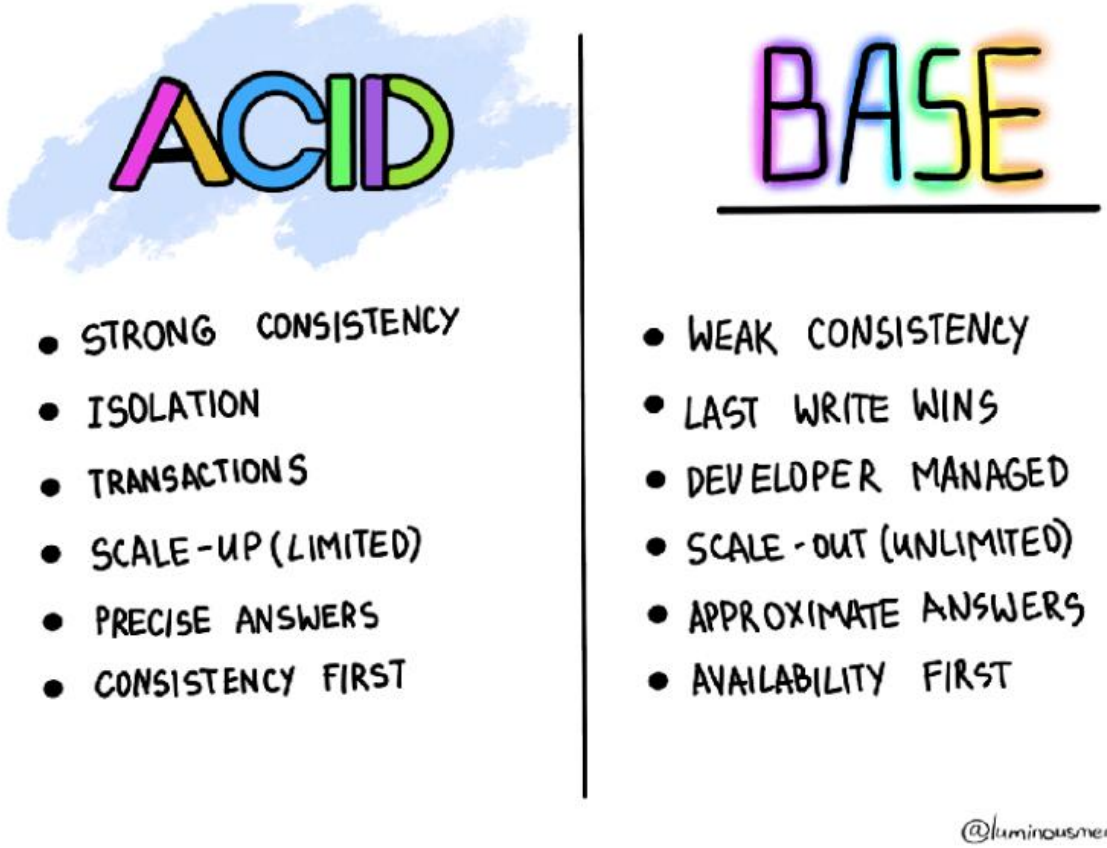


Database Classification



ACID vs BASE

- Acronyms describing database behavior and characteristics
- Most SQL databases implement ACID, and most No-SQL databases implement BASE. However, there are exceptions!
- ACID: Atomicity, consistency, isolation, and durability.
- BASE: Basically available, soft state, and eventually consistent.



https://aws.amazon.com/de/compare/the-difference-between-acid-and-base-database/

Relational Databases

Use is advisable when

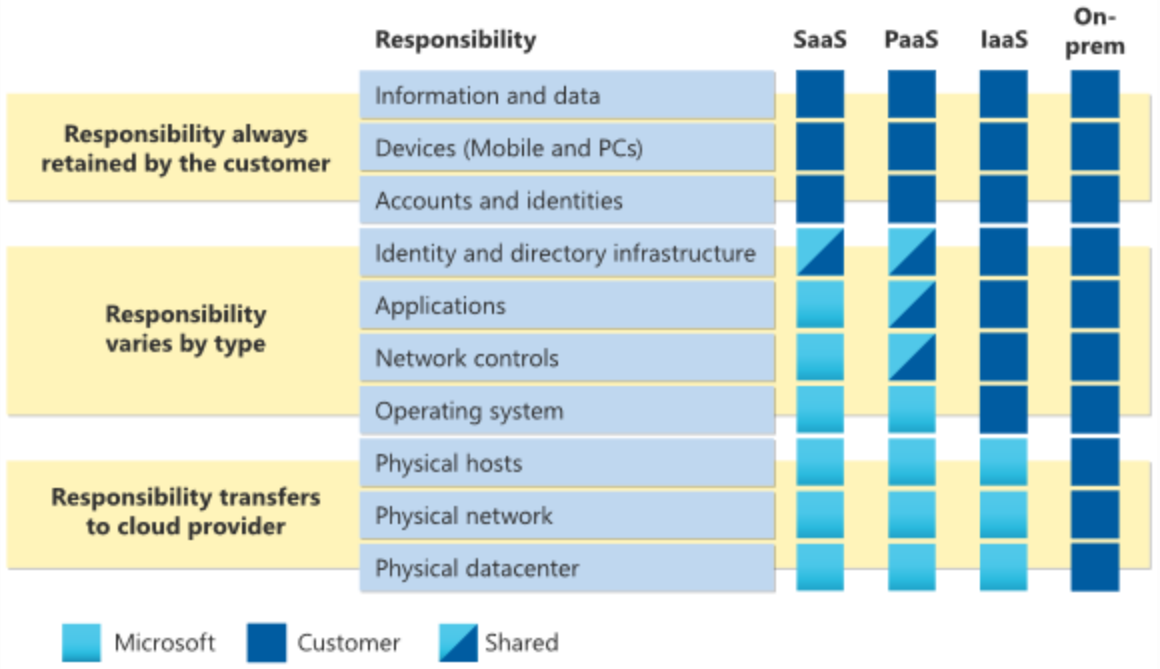
- Data schema is specified and must enforce rules (data quality)
- ACID necessary
- Number of read and write operations is limited
- Moderate performance necessary

Benefits

- User-friendliness
- Data integrity
- Low storage requirements
- SQL – Structured Query Language

Hosting on Azure

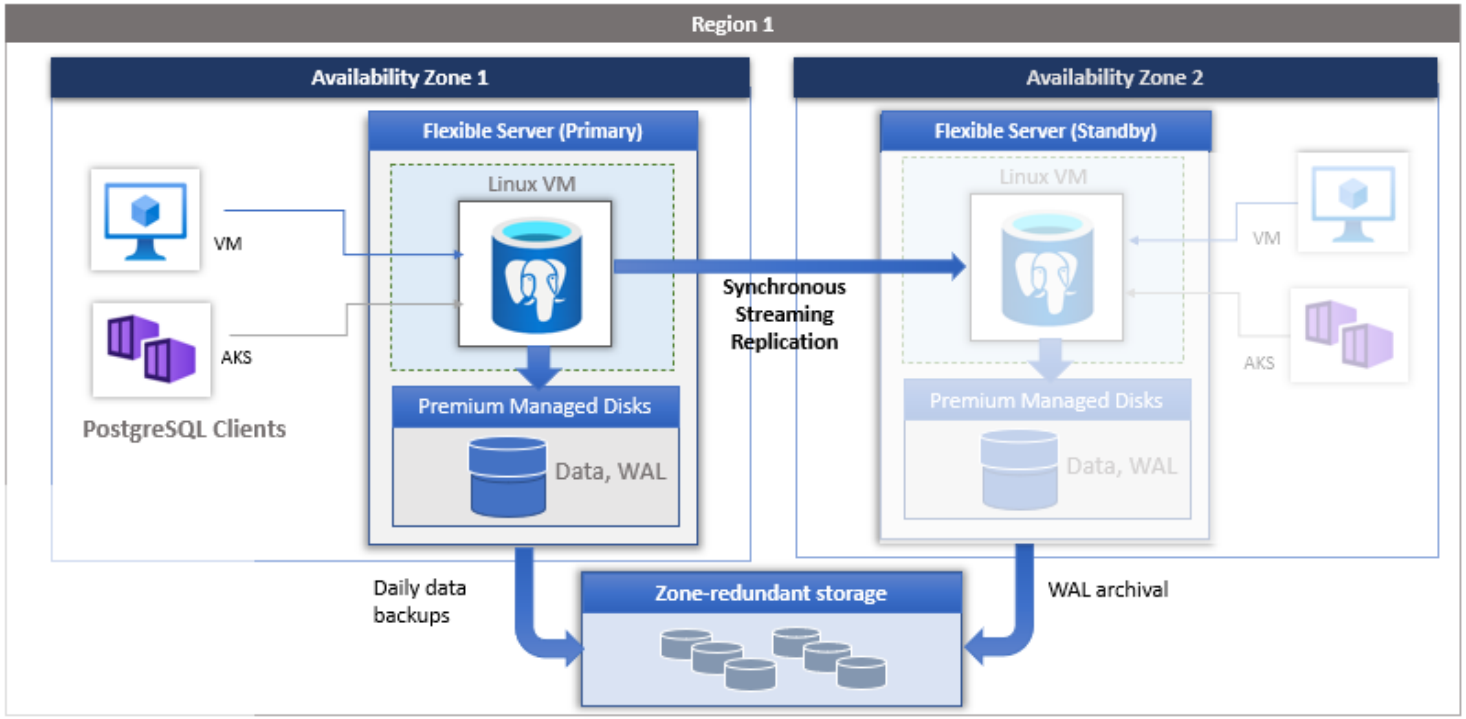
- IaaS: Self-managed IaaS VMs with, for example, SQL Server on top.
  - Customer bears overall responsibility.
- DBaaS: Managed database servers, e.g., Managed Postgresql.
  - Dedicated VMs for the instances.
  - Operation by the provider.
  - Personal responsibility for, e.g., backup and restore processes, data residency, and availability levels.



https://learn.microsoft.com/en-us/azure/security/fundamentals/shared-responsibility

Azure Database for PostgreSQL Flexible Server

- Managed PosgreSQL instance from Azure
- Configuration of
  - Data residency (region)
  - Replication (read replicas)
  - Backup intervals
  - Encryption
  - Network integration (private endpoint or similar)



Azure Database for PostgreSQL Flexible Server SKUs

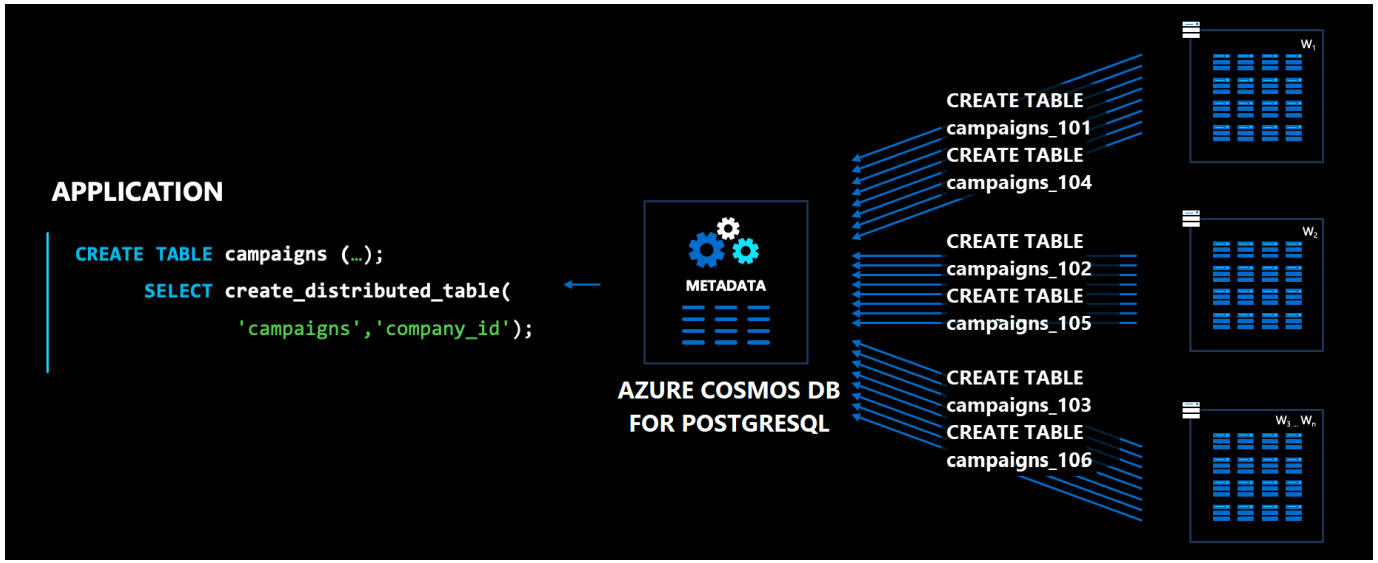
Resource/Tier	Burstable	General Purpose	Memory Optimized
VM-series	B-series	Ddsv5-series, Dadsv5-series, Ddsv4-series, Dsv3-series	Edsv5-series, Eadsv5-series, Edsv4-series, Esv3-series
vCores	1, 2, 4, 8, 12, 16, 20	2, 4, 8, 16, 32, 48, 64, 96	2, 4, 8, 16, 20 (v4/v5), 32, 48, 64, 96
Memory per vCore	Variable	4 GiB	6.75 GiB to 8 GiB
Storage size	32 GiB to 64 TiB	32 GiB to 64 TiB	32 GiB to 64 TiB
Automated Database backup retention period	7 to 35 days	7 to 35 days	7 to 35 days
Long term Database backup retention period	up to 10 years	up to 10 years	up to 10 years

https://learn.microsoft.com/en-us/azure/postgresql/flexible-server/concepts-compute

Azure Cosmos DB for PostgreSQL

Azure Cosmos DB is a

- Fully managed DB service with various DB APIs (e.g., PostgreSQL, MongoDB, Apache Cassandra, and more)
- Optional configuration as a multi-region and multi-master service instance
- Features of specific APIs may not always correspond fully to the product (e.g., in the case of MongoDB)
- Hint: Cosmos DB for PostgreSQL is followed by newer services



https://learn.microsoft.com/en-us/azure/cosmos-db/postgresql/introduction und https://learn.microsoft.com/en-us/azure/cosmos-db/introduction

NoSQL Datenbanken

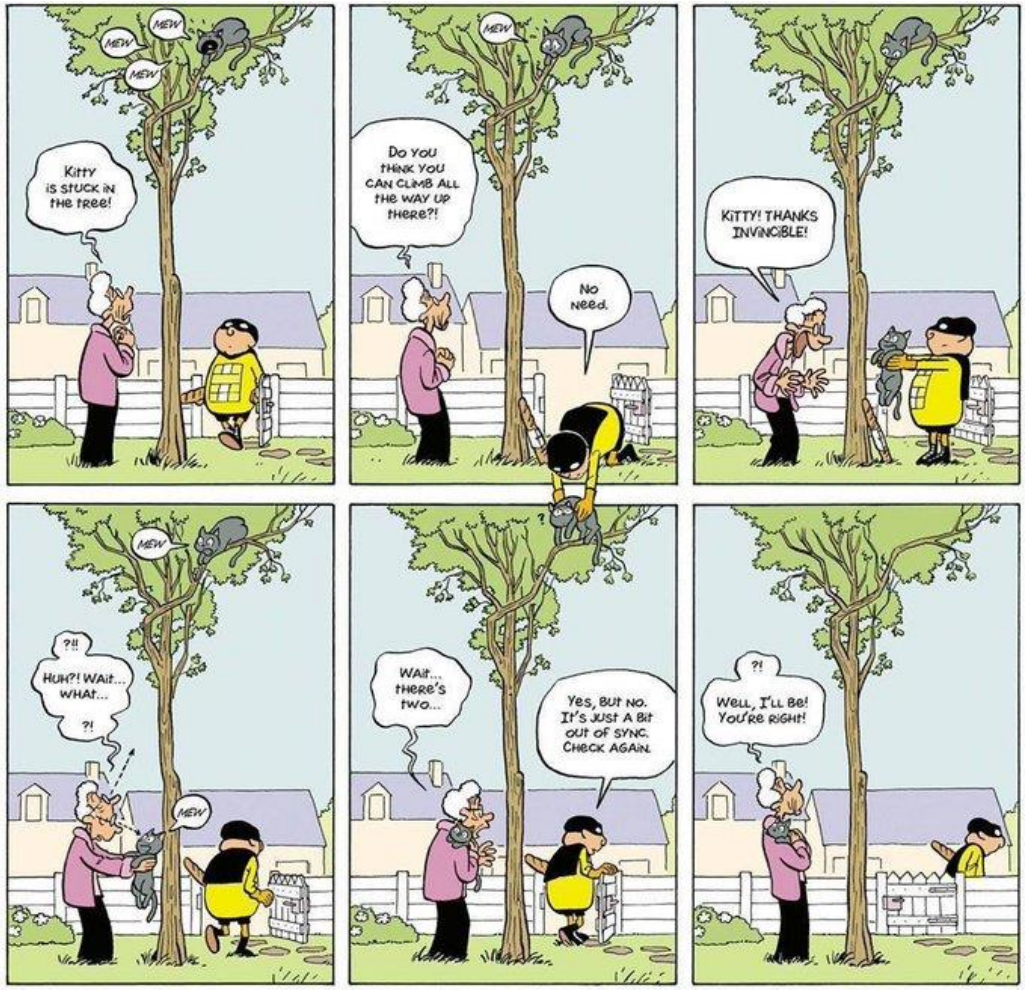
Use is advisable when

- Processing of high data volumes
- Horizontal scalability required (e.g., across regions)
- No fixed schemas for the data
- High volume of read and write operations

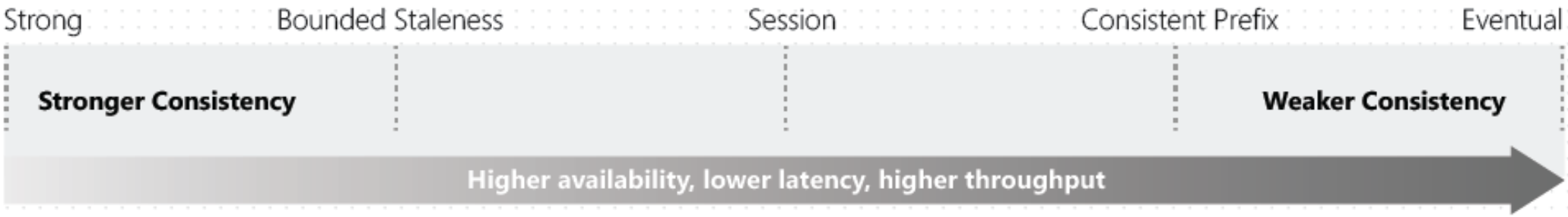
Benefits

- Flexibility
- Scalability
- High performance
- APIs

Eventual Consistency



Azure Cosmos DB Eventual Consistency



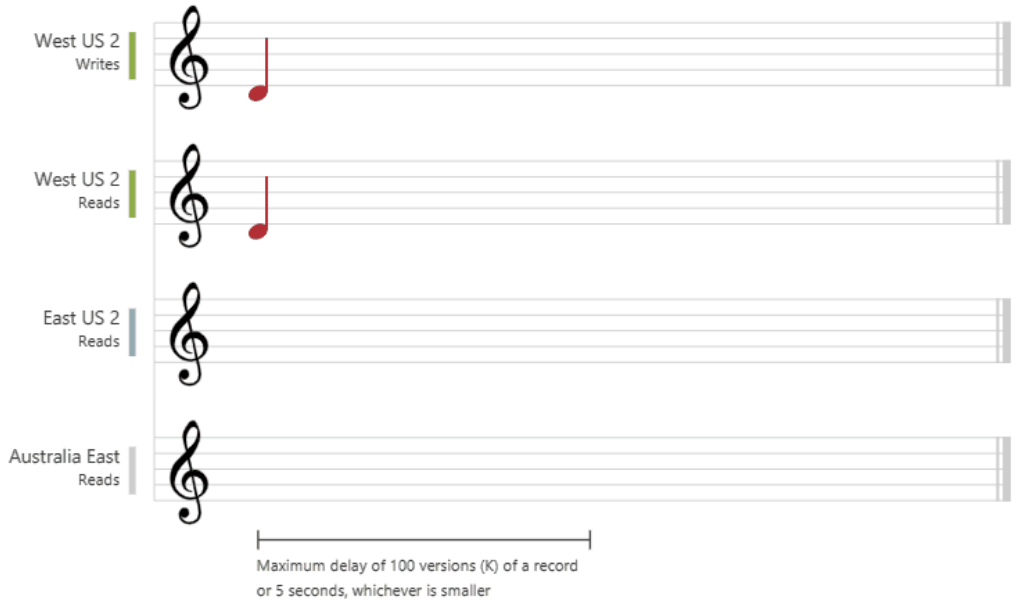
- Cosmos DB offers 5 consistency levels
- From left to right
  - Data must first be committed in all replicas vs. new data record is replicated step by step (e.g., across replicas and regions)
  - The latest version of the data record is read vs. an outdated state may be read

https://learn.microsoft.com/en-us/azure/cosmos-db/consistency-levels

Azure Cosmos DB Eventual Consistency

Strong consistency

Bounded staleness

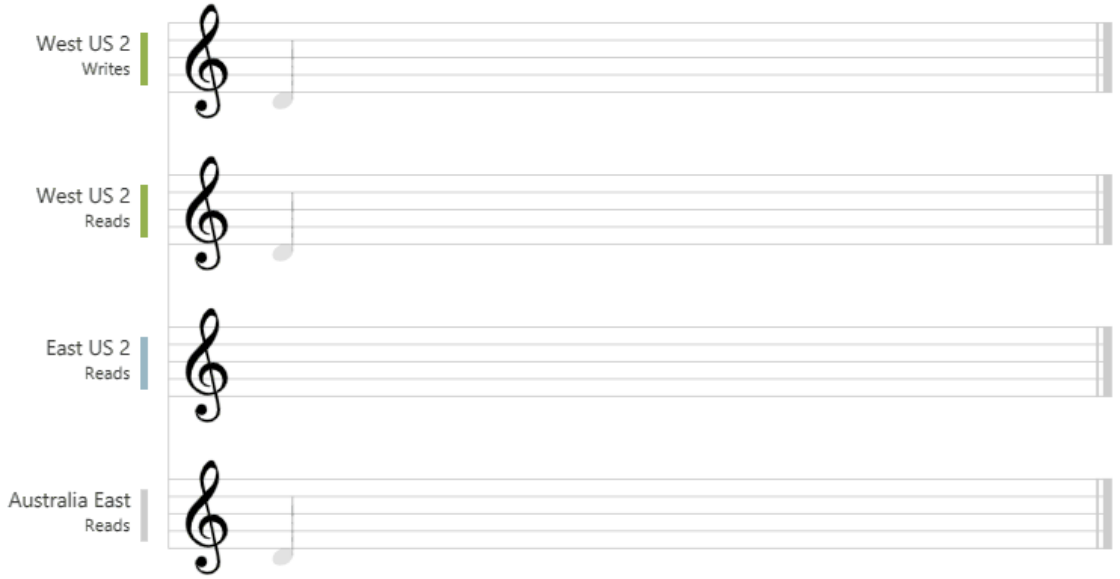


https://learn.microsoft.com/en-us/azure/cosmos-db/consistency-levels

Azure Cosmos DB Eventual Consistency

Session consistency

Consistent prefix consistency



https://learn.microsoft.com/en-us/azure/cosmos-db/consistency-levels

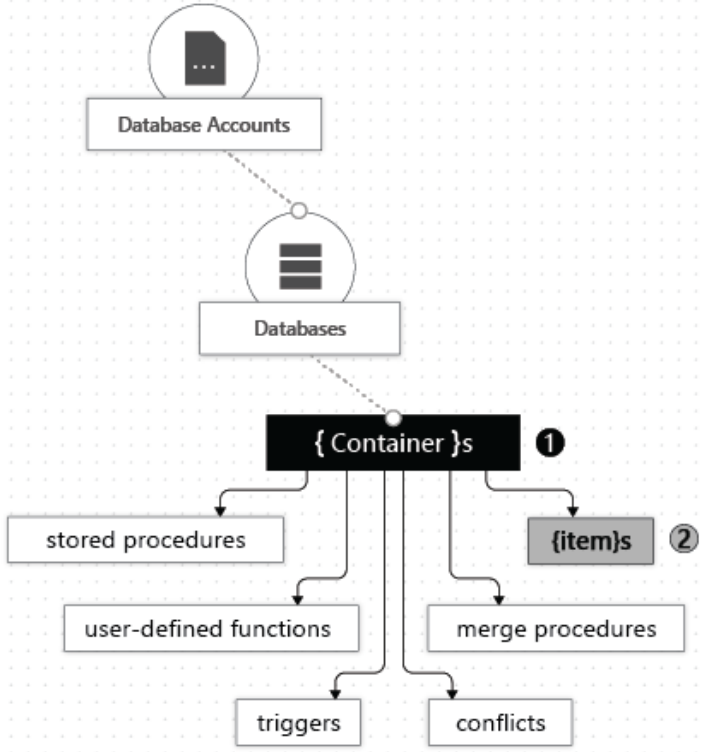
Azure Cosmos DB Eventual Consistency

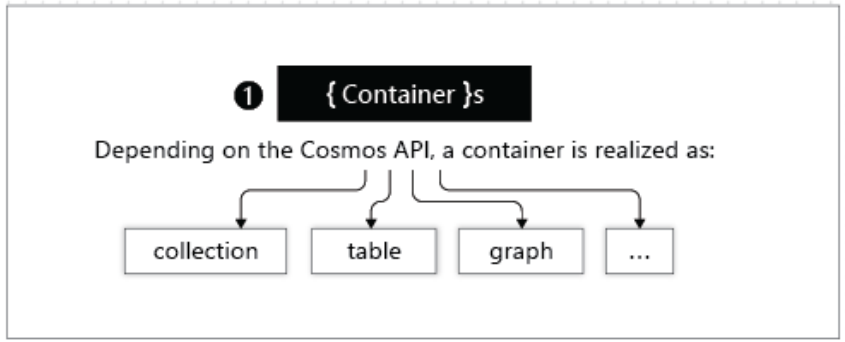
- Eventual consistency is useful for (globally) distributed applications with high performance and availability requirements.
- The consistency level must match the use case (outdated data may be displayed or needs to be handled by the app).
- Consistency levels can be selected for each API. Different defaults are used.

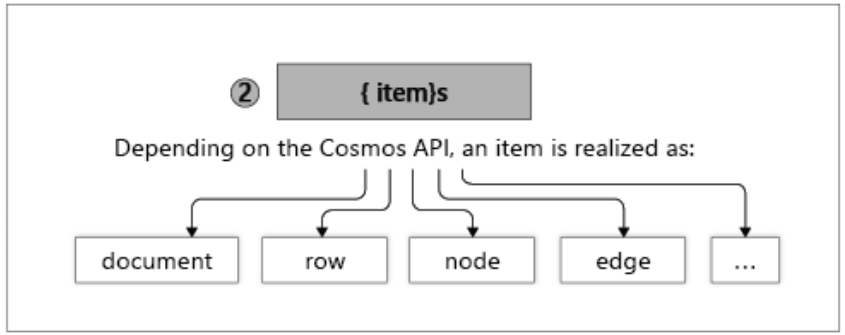
https://learn.microsoft.com/en-us/azure/cosmos-db/consistency-levels

Cosmos DB No-SQL

- The generic model of Cosmos DB is mapped to the specific APIs
- Example DocumentDB:
  - Container = Collection
  - Item = Document



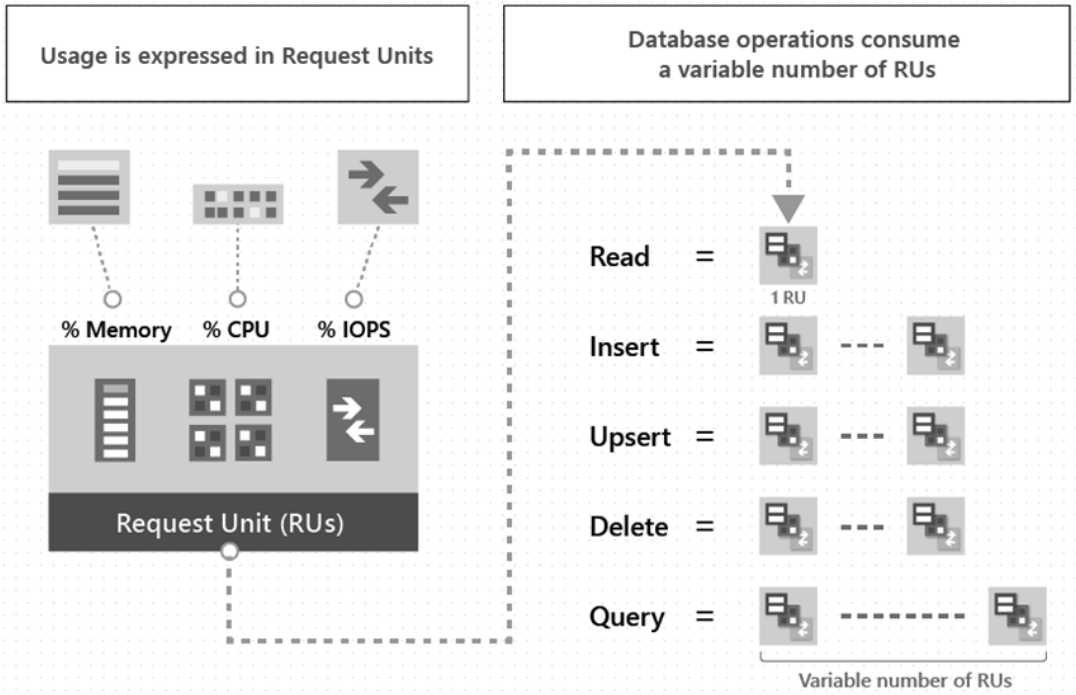




https://learn.microsoft.com/en-us/azure/cosmos-db/resource-model?context=%2Fazure%2Fcosmos-db%2Fnosql%2Fcontext%2Fcontext

Cosmos DB - Pricing

- Compute Pricing
  - Request Units (RUs) – a generic unit for mapping CPU and memory
  - Alternatively, vCores
- Storage Pricing
  - Used storage (GB)
  - Alternatively, provisioned storage (disk size per Cosmos DB node)
- Resources are obtained via
  - Serverless SKU (effective consumption)
  - Provisioned throughput/storage (fixed amount of RUs or vCores)



https://learn.microsoft.com/en-us/azure/cosmos-db/request-units

Cosmos DB – Postgres API

- Cosmos DB instance with Posgres API compatibility
- vCore based deployment
- Coordinator: Contact point for clients, forwards queries to nodes

```
@secure()
param administratorLoginPassword string
param location string
param clusterName string
param coordinatorVCores int = 4
param coordinatorStorageQuotaInMb int = 262144
param coordinatorServerEdition string = 'GeneralPurpose'
param enableShardsOnCoordinator bool = true
param nodeServerEdition string = 'MemoryOptimized'
param nodeVCores int = 4
param nodeStorageQuotaInMb int = 524288
param nodeCount int
param enableHa bool
param coordinatorEnablePublicIpAccess bool = true
param nodeEnablePublicIpAccess bool = true
param availabilityZone string = '1'
param postgresqlVersion string = '15'
param citusVersion string = '12.0'

resource serverName_resource 'Microsoft.DBforPostgreSQL/serverGroupsv2@2022-11-08'
        name: clusterName
        location: location
        tags: {}
        properties: {
                administratorLoginPassword: administratorLoginPassword
                coordinatorServerEdition: coordinatorServerEdition
                coordinatorVCores: coordinatorVCores
                coordinatorStorageQuotaInMb: coordinatorStorageQuotaInMb
                enableShardsOnCoordinator: enableShardsOnCoordinator
                nodeCount: nodeCount
                nodeServerEdition: nodeServerEdition
                nodeVCores: nodeVCores
                nodeStorageQuotaInMb: nodeStorageQuotaInMb
                enableHa: enableHa
                coordinatorEnablePublicIpAccess: coordinatorEnablePublicIpAccess
                nodeEnablePublicIpAccess: nodeEnablePublicIpAccess
                citusVersion: citusVersion
                postgresqlVersion: postgresqlVersion
                preferredPrimaryZone: availabilityZone
                }
        }
```

https://learn.microsoft.com/en-us/azure/cosmos-db/postgresql/quickstart-create-bicep?tabs=CLI

Cosmos DB – SQL API

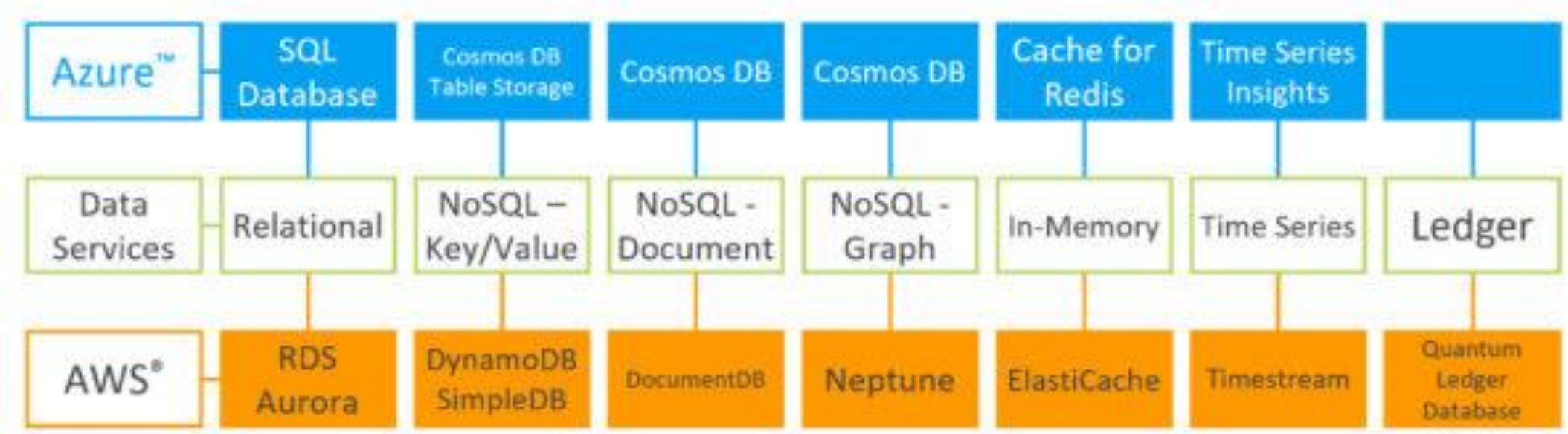
```
resource account 'Microsoft.DocumentDB/databaseAccounts@2023-11-15' = {
  name: toLower(accountName)
  location: location
  properties: {
    enableFreeTier: true
    databaseAccountOfferType: 'Standard'
    consistencyPolicy: {
      defaultConsistencyLevel: 'Session'
    }
    locations: [
      {
        locationName: location
      }
    ]
  }
}

resource database 'Microsoft.DocumentDB/databaseAccounts/sqlDatabases@2023-11-15'
  parent: account
  name: databaseName
  properties: {
    resource: {
      id: databaseName
    }
    options: {
      throughput: 1000
    }
  }
}
```

```
resource container 'Microsoft.DocumentDB/databaseAccounts/sqlDatabases/containers@
  parent: database
  name: containerName
  properties: {
    resource: {
      id: containerName
      partitionKey: {
        paths: [
          '/myPartitionKey'
        ]
        kind: 'Hash'
      }
      indexingPolicy: {
        indexingMode: 'consistent'
        includedPaths: [
          {
            path: '/*'
          }
        ]
        excludedPaths: [
          {
            path: '/_etag/?'
          }
        ]
      }
    }
  }
}
```

https://learn.microsoft.com/en-us/azure/cosmos-db/nosql/manage-with-bicep#free-tier-azure-cosmos-db-account

Summary DB Services



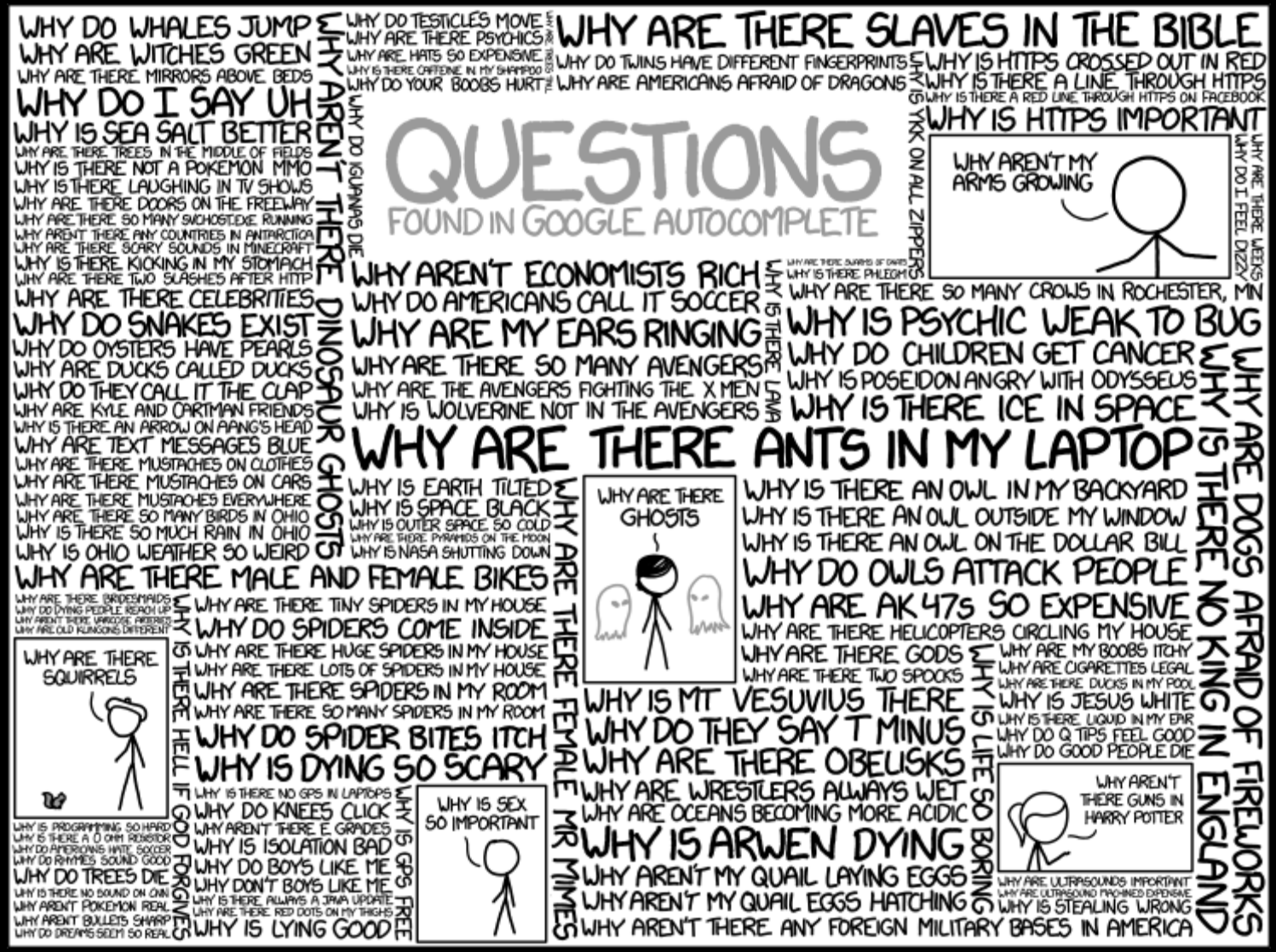
Demo: CosmosDB with Functions

Databases - Summary

- Relational databases are primarily based on ACID, whereas most No-SQL databases are based on BASE.
- Database as a Service transfers OS and database maintenance tasks to the cloud provider.
- Cost savings in database operation are offset via OPEX.
- Existing database technologies and versions may vary depending on the region and cloud provider.

Hausaufgaben

- Review the material from this lesson.
- Complete Assignment 09.
- Create and deploy your own AI service instances with the following learning paths (if needed):
  - Get started with Azure Cosmos DB for NoSQL
  - Azure Database for PostgreSQL Tutorials


QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY ARE THERE SLAVES IN THE BIBLE
WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SVCHOST.EXE RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REACH UP
WHY AREN'T THERE VARICOSE ARTERIES
WHY ARE OLD KLINGONS DIFFERENT
WHY ARE THERE SQUIRRELS
WHY IS PROGRAMMING SO HARD
WHY IS THERE A 0 OHM RESISTOR
WHY DO AMERICANS HATE SOCCER
WHY DO RHYMES SOUND GOOD
WHY DO TREES DIE
WHY IS THERE NO SOUND ON CNN
WHY AREN'T POKEMON REAL
WHY AREN'T BULLETS SHARP
WHY DO DREAMS SEEM SO REAL
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY THIGHS
WHY IS LYING GOOD
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY IS HTTPS IMPORTANT
WHY AREN'T MY ARMS GROWING
WHY ARE THERE SWARMS OF DAVES
WHY IS THERE PHLEGM
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY ARE THERE ANTS IN MY LAPTOP
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE GHOSTS
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOOKS
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND MACHINES EXPENSIVE
WHY IS STEALING WRONG
WHY IS SEX SO IMPORTANT
WHY DO IGUANAS DIE
WHY AREN'T THERE DINOSAUR GHOSTS
WHY IS THERE LAVA
WHY ARE THERE FEMALE MR MIMES
WHY IS GPS FREE
WHY IS LIFE SO BORING
WHY IS THERE NO KING IN ENGLAND
WHY ARE DOGS AFRAID OF FIREWORKS
WHY DO I FEEL DIZZY
WHY ARE THERE WEEKS
WHY IS YYK ON ALL ZIPPERS
WHY DO FORGIVES